

ORBIFOLD AND LOGARITHMIC IITAKA SUBADDITIVITY

OPENAI

ABSTRACT. We prove Campana’s orbifold Iitaka subadditivity conjecture for rational simple normal crossing boundaries on compact manifolds in Fujiki class $\mathcal{C}$, including coefficient one. Ordinary and logarithmic subadditivity follow.

CONTENTS

1. Introduction	1
2. The relative Iitaka reduction and its Hodge line	5
3. The adjoint-addition principle	17
4. Adjoint positivity of an extreme Hodge line	20
5. Addition with a log-general-type fiber	31
6. Return to the original fibration and consequences	39
7. Minimal root covers for reduced boundaries	43
Appendix A. Supplementary projective calculations	48
References	53

1. INTRODUCTION

The Kodaira dimension measures the growth of pluricanonical forms. For a surjective morphism $f : X \rightarrow Y$ of smooth projective complex varieties with connected fibers, Iitaka’s subadditivity problem asks whether

$$\kappa(X) \geq \kappa(F) + \kappa(Y),$$

where F is a general fiber. The logarithmic version permits poles along reduced boundary divisors and therefore applies to open varieties. We prove a common generalization in which rational boundary coefficients and multiple fibers both contribute to the inequality.

Iitaka’s divisor-dimension and logarithmic theories provide the language for this question [40, 41]; an explicit logarithmic subadditivity conjecture appears in [42, §3]. The lower bound requires global pluricanonical forms that retain the growth of both the fibers and the base. Direct-image positivity established central cases: Kawamata treated curve bases and fibers admitting good minimal models, Viehweg developed weak positivity and addition over a base of general type, and Kollár proved addition for general-type fibers [43, 44, 60, 45]. Weak positivity controls symmetric powers after small positive ample twists; a general-type base provides enough canonical positivity to absorb them.

Other advances exploit dimension or the geometry of the base: Birkar treated total dimension at most six, Cao–Păun and Hacon–Popa–Schnell treated abelian or maximal-Albanese-dimension bases, and Cao treated projective klt pairs over surfaces [10, 18, 37, 17]. Thus the established cases draw positivity from different parts of the fibration, rather than imposing one uniform condition on it.

For logarithmic pairs, Kovács–Patakfalvi proved the projective log-general-type-fiber case [46, Theorem 9.6], while Hashizume proved reduced-SNC subadditivity for fibers with abundant log canonical divisor [38, Theorem 1.2]. Maehara had established logarithmic addition over a base of log general type; Fujino gives a proof through twisted weak positivity [27, Theorem 1.9]. Campana proved addition over an orbifold base of general type, and Wang proved klt Kähler subadditivity over complex tori, including an inf-multiplicity refinement [16, Theorem 6.3] [62, Main Theorem A and Theorem E]. The present argument uses general-type addition only after a relative Iitaka reduction; the original fiber need not be of general type or admit a good minimal model.

Earlier preprints also state general ordinary inequalities. Tsuji states ordinary subadditivity following a direct-image generation theorem [58, Theorems 1.9 and 1.13]; his Remark 1.14 distinguishes an unproved variation refinement. Maehara states an ordinary determinant inequality involving variation [47, §7, Theorem 8]. This later preprint claim is distinct from his logarithmic general-type-base theorem above. Neither preprint statement is a proof input here.

1.1. The orbifold base and the main theorem. Multiple fibers carry information that the ordinary base omits. Campana’s orbifold base records it through inf multiplicities, and his birational invariant gives the corresponding form of Iitaka’s conjecture [15, 16]. We specify that invariant before stating the theorem.

A fibration is a proper surjective holomorphic map with connected fibers. A rational SNC boundary on a smooth manifold is a finite sum $\Delta = \sum_i \delta_i \Delta_i$, with $\delta_i \in \mathbf{Q} \cap [0, 1]$ and simple normal crossing support. Write Δ_E for the coefficient of a source prime E . Set

$$m_\Delta(E) = \frac{1}{1 - \Delta_E}, \quad \frac{1}{0} = \infty,$$

where $\Delta_E = 0$ off the boundary. If the base of $f : (X, \Delta) \rightarrow Y$ is smooth, define

$$(1.1) \quad m(f, \Delta; D) = \min_{E \rightarrow D} \{\text{ord}_E(f^* D) m_\Delta(E)\}, \quad B(f, \Delta) = \sum_D \left(1 - \frac{1}{m(f, \Delta; D)}\right) D.$$

Here D ranges over base prime divisors, E over source primes dominating D , and $1/\infty = 0$. The minimum is nonempty and only finitely many coefficients of $B(f, \Delta)$ are nonzero. There is no divisibility condition on these multiplicities.

For smooth sources and bases, an elementary change of model is a commutative diagram of fibrations

$$\begin{array}{ccc} (X', \Delta') & \xrightarrow{u} & (X, \Delta) \\ f' \downarrow & & \downarrow f \\ Y' & \xrightarrow{v} & Y \end{array}$$

where u, v are proper bimeromorphic holomorphic maps, $u_* \Delta' = \Delta$, and both boundaries are rational SNC boundaries in $[0, 1]$. We require, for every prime D on X and prime E on X' with $t = \text{ord}_E(u^* D) > 0$,

$$(1.2) \quad t m_{\Delta'}(E) \geq m_\Delta(D).$$

We permit $\infty \geq \infty$ and impose no condition for $t = 0$. Write $f' \sim f$ for the equivalence relation generated by these diagrams, allowing arrows in either direction. Define

$$(1.3) \quad \kappa(f \mid \Delta) = \inf_{f' : (X', \Delta') \rightarrow Y' \sim f} \kappa(Y', K_{Y'} + B(f', \Delta')).$$

For a normal singular base Y , resolve Y , resolve the main component of the fiber product with X , and give the resulting smooth source the strict transform of Δ plus the reduced exceptional

divisor over X . After resolving this boundary, apply the preceding definition. It is independent of the chosen resolution [16, Definitions 2.1, 2.3, 3.2, 4.7, and 4.10; Remark 4.13 and Corollary 4.14].

For a rational line bundle L , $\kappa(Z, L)$ is the maximal image dimension of its complete linear systems in sufficiently divisible positive degrees, and is $-\infty$ if all these systems are empty. A point has Kodaira dimension zero, and $(-\infty) + a = -\infty$, including $a = -\infty$. Fujiki class $\mathcal{C}$ consists of compact complex spaces bimeromorphic to compact Kähler spaces.

Theorem 1.1 (Orbifold subadditivity). *Let X be a smooth compact connected complex manifold in Fujiki class $\mathcal{C}$, let Δ be a rational SNC boundary, and let $f : X \rightarrow Y$ be a surjective holomorphic map with connected fibers onto a normal compact irreducible complex space. For a very general smooth fiber F , put $\Delta_F = \Delta|_F$. Then*

$$(1.4) \quad \kappa(X, K_X + \Delta) \geq \kappa(F, K_F + \Delta_F) + \kappa(f|_F).$$

This is a positive resolution of Campana's orbifold Iitaka conjecture in its rational SNC, Fujiki-class- $\mathcal{C}$ formulation, including coefficient one [16, Conjecture 6.1 and Remark 6.2]. Here a very general fiber is chosen outside a countable union of proper closed analytic subsets, including the critical values and the exceptional images of boundary strata.

For reduced SNC divisors D_X, D_Y satisfying $\text{Supp}(f^*D_Y) \subseteq \text{Supp}(D_X)$, the invariant base term is at least $\kappa(Y, K_Y + D_Y)$. Theorem 1.1 therefore gives

$$\kappa(X, K_X + D_X) \geq \kappa(F, K_F + D_X|_F) + \kappa(Y, K_Y + D_Y)$$

for fibrations between smooth compact class- $\mathcal{C}$ manifolds. For a smooth quasi-projective variety U , its logarithmic Kodaira dimension is $\bar{\kappa}(U) = \kappa(X, K_X + D_X)$ on a smooth projective compactification with reduced SNC boundary $D_X = X \setminus U$. Compatible compactifications give logarithmic subadditivity for dominant morphisms of smooth quasi-projective complex varieties with geometrically connected general fiber. Empty boundaries give ordinary subadditivity. The projective ordinary and logarithmic statements also hold over algebraically closed fields of characteristic zero, using geometric generic fibers.

1.2. The reduction and the two positivity inputs. The proof first replaces the original fiber by a general-type object without losing its complete pluricanonical systems. Assume the two terms on the right of Theorem 1.1 are nonnegative, and put $k = \kappa(F, K_F + \Delta_F)$. A relative Iitaka map gives, after the model changes of Section 2,

$$X \xrightarrow{g} W \xrightarrow{h} Y, \quad \dim W_y = k, \quad \kappa(G, K_G + \Delta_G) = 0$$

for a very general g -fiber G .

The canonical-bundle-formula strategy separates the contribution of singular fibers from a moduli contribution on W . Fujino–Mori developed a higher-dimensional logarithmic formula, and Ambro organized its discriminant and moduli parts birationally [32, 2, 3]. Our use of this strategy requires an exact comparison on the fixed original source. Section 2 obtains it from the orders of relative pluriforms. A sufficiently divisible pluricanonical system on G is one-dimensional. Taking a root of its generator on a cyclic cover gives a differential form, and the line it spans belongs to the cohomology of that cover. Tensor descent and extension across the boundary produce a rational line bundle M on W . The orders of the same form determine a boundary T such that

$$B(g, \Delta) \leq T \leq 1, \quad H^0(X, m(K_X + \Delta)) \simeq H^0(W, m(K_W + T + M))$$

in sufficiently divisible degrees. The comparison also holds relatively over Y . Thus $K_W + T + M$ is big on the k -dimensional fibers of h .

Two inputs turn this fiberwise bigness into the required global lower bound. The first concerns the line M : the period map, which records how the Hodge structures of the covers vary, supplies

a second fibration $p : W \rightarrow S$, with S smooth projective, and a nef rational line bundle P such that

$$M = p^*P, \quad K_S + aP \text{ is big}$$

for some positive rational a . The maps h and p have different jobs, as Figure 1 shows. The second input is log-general-type addition: a log canonical divisor that is big on the fibers of h has Kodaira dimension at least their dimension plus the orbifold contribution of Y .

$$\begin{array}{ccccc} X & \xrightarrow{g} & W & \xrightarrow{h} & Y \\ & & \downarrow p & & \\ & & S & & \end{array}$$

FIGURE 1. The relative Iitaka reduction factors the original fibration through W . The map h retains the original base and has k -dimensional general fibers. The independent map p expresses M as the pullback of a nef line P on a projective base where $K_S + aP$ is big. No map between Y and S is asserted.

Section 3 states both inputs precisely and proves the intervening addition principle. Write $D_W = K_W + T$. For a fixed rational $c < 1$ close to one, general-type addition gives sufficiently many ratios of sections of $D_W + cM + \eta p^*A$, where A is ample on S and $\eta > 0$ is rational. Weak positivity then supplies one fixed section of $D_W + aM - \lambda p^*A$ for suitable $a > 1$ and $\lambda > 0$. A positive rational combination cancels the ample terms and gives coefficient one on M . Multiplication by powers of that fixed section preserves the ratios of the first system. This is an exact section argument; it requires neither abundance of P nor a limiting assertion for Kodaira dimensions.

The full proofs of the inputs follow this calculation. Section 4 proves adjoint positivity of the Hodge line. It combines the curvature and boundary theory of Griffiths, Schmid, and Cattani–Kaplan–Schmid with a projective period quotient [35, 56, 19]. The construction retains suitable rational adjoint factors of the ambient integral variation, associated with its monodromy, from which a positive power of the chosen line can be recovered. Their full periods give the quotient, using Villadsen’s class- $\mathcal{C}$ compactification theorem and the period-quotient approach of Bakker–Brunebarbe–Tsimmerman and Sommesse; curvature and a volume argument establish its projectivity [6, 57, 61].

On this quotient S , the retained full period map is generically immersive, but the map remembering only the chosen line may have smaller rank. Curvature identifies the latter rank with the numerical dimension $\nu(P)$, measured by its nonzero intersection powers. Let D be the reduced boundary where the retained variation has nonidentity local monodromy. Brunebarbe–Cadorel gives $K_S + D$ big [14]. Using André’s monodromy normality and Bakker–Tsimmerman’s Ax–Schanuel theorem to analyze fibers of the line map, the boundary argument proves $\nu(P|_{D_i}) < \nu(P)$ for every component D_i of D , including finite nonidentity monodromy [4, 8]. The boundary loss in the asymptotic section count consequently has lower degree in the coefficient of P than the available sections on S . Lemma 4.7 turns this comparison into bigness of $K_S + aP$.

Section 5 proves log-general-type addition in class $\mathcal{C}$. Kovács–Patakfalvi’s stable-family positivity applies to projective families. We construct such an auxiliary family and descend its relative sections to the original class- $\mathcal{C}$ base. Fujino’s weak positivity supplies the other direct-image input used in Section 3 [46, 27]. Section 6 then returns once to the original fibration, applies the addition principle to (W, T, M) , and derives the ordinary, logarithmic, characteristic-zero, and core consequences. The core and the relevant specialness of Kodaira-zero fibers are Campana’s constructions and results [15, 16].

1.3. The reduced-boundary refinement. For a rational boundary, the root form above spans one character subspace of the cyclic cover's highest Hodge space; the whole highest space can be larger. For projective reduced-boundary pairs of logarithmic Kodaira dimension zero, Section 7 proves more: the minimal root cover has a one-dimensional entire logarithmic top-form space. Mixed-Hodge extension and semipositivity in the forms of Fujino–Fujisawa control the coefficient-one boundary [29, 30]. We identify the resulting line with the moduli divisor in the projective subpair setting of Bakker–Filipazzi–Mauri–Tsimmerman [7, Theorem 6.28]. This identification does not use their separate semiample theorem, whose hypotheses include effectiveness over the generic point [7, Theorem 1.5]. Section 7 also gives the exact divisorial normalization. Appendix A collects complementary projective calculations by valuations and toroidal models, together with the comparison on higher models.

The logarithmic lower bound, adjoint positivity, and this reduced-boundary refinement supply the stated inputs to the whole-fiber-variation companion [50]. The reverse-inequality companion [52] uses the lower bound only for additivity; its upper theorem has additional smoothness hypotheses on all boundary strata. Neither companion is used in the proof of Theorem 1.1. The separate article on projective Hodge lines [51] gives a complete ordinary proof through rationally polarized integral variations, together with the distinct integral and analytic boundary arguments.

2. THE RELATIVE IITAKA REDUCTION AND ITS HODGE LINE

The first task is to replace the fibers by their Iitaka images while retaining every sufficiently divisible pluricanonical system. We first fix the birational models on which divisorial order tests suffice. The root form of a Kodaira-zero fiber will then provide a Hodge line and an exact comparison of sections on the intermediate base.

We use additive notation for rational line bundles; equality of them means equality in $\text{Pic} \otimes \mathbf{Q}$. A rational line is effective if a positive integral multiple has a nonzero section. All degree assertions below are made after clearing denominators. For a fibration $g : (Z, \Gamma) \rightarrow T$ and a very general smooth fiber F , adjunction gives

$$(2.1) \quad (K_Z + \Gamma)|_F = K_F + \Gamma|_F.$$

2.1. The fixed smooth models. A smooth-base fibration $g : (Z, \Gamma) \rightarrow T$ is *neat* if there is a proper bimeromorphic orbifold morphism $w : (Z, \Gamma) \rightarrow (Z_0, \Gamma_0)$ to a smooth pair, with $w_*\Gamma = \Gamma_0$, such that every prime divisor sent by g into codimension at least two is also sent by w into codimension at least two. Campana's construction gives suitable neat models, model independence, and the formula

$$(2.2) \quad \kappa(g | \Gamma) = \kappa(T, K_T + B(g, \Gamma)) \quad \text{on a neat model}$$

[16, Definition 3.8, Proposition 3.10, Corollary 4.11, and Remark 6.2]. For the proof we only need the defining inequality, valid on every allowed smooth model:

$$(2.3) \quad \kappa(T, K_T + B(g, \Gamma)) \geq \kappa(g | \Gamma).$$

2.2. Reduced exceptional boundaries.

Lemma 2.1 (Logarithmic modifications). *Let $p : Z' \rightarrow Z$ be a proper bimeromorphic holomorphic map between smooth compact manifolds. Suppose that Γ has simple normal crossing support and coefficients in $\mathbf{Q} \cap [0, 1]$, and set*

$$\Gamma' = p_*^{-1}\Gamma + \text{Exc}(p)_{\text{red}}.$$

Assume, after further resolution if necessary, that this boundary has simple normal crossing support. There is an effective p -exceptional rational divisor A such that

$$(2.4) \quad K_{Z'} + \Gamma' = p^*(K_Z + \Gamma) + A.$$

For every sufficiently divisible positive integer m , the natural identification of meromorphic pluricanonical forms induces

$$(2.5) \quad p_* \mathcal{O}_{Z'}(m(K_{Z'} + \Gamma')) = \mathcal{O}_Z(m(K_Z + \Gamma)).$$

These identifications respect multiplication, so the divisible section rings and their Iitaka dimensions agree. Moreover, p satisfies (1.2).

Proof. The difference in (2.4) is exceptional, since p is an isomorphism at the generic point of every target prime. To compute its coefficient at an exceptional prime E , choose compatible local canonical forms. Near a general point of its center on Z , let $z_1, \dots, z_n$ be coordinates in which the boundary components containing that center are $z_i = 0$ for $1 \leq i \leq r$, with coefficients γ_i . Put $t_i = \text{ord}_E(p^* z_i)$ and let k_E be the order of the Jacobian of p along E . The pullback of

$$\frac{dz_1}{z_1} \wedge \dots \wedge \frac{dz_r}{z_r} \wedge dz_{r+1} \wedge \dots \wedge dz_n$$

has at most a simple pole along E . Indeed, at a general smooth point of E , each logarithmic factor has the form $t_i du/u$ plus a holomorphic one-form, and a nonzero wedge product contains at most one factor du/u . Consequently $k_E + 1 \geq \sum_i t_i$. The coefficient in question is therefore

$$k_E + 1 - \sum_i \gamma_i t_i \geq \sum_i (1 - \gamma_i) t_i \geq 0.$$

For an effective integral exceptional divisor mA , normality of Z gives $p_* \mathcal{O}_{Z'}(mA) = \mathcal{O}_Z$: a meromorphic function with poles only on mA descends outside the codimension-two exceptional image and extends by Hartogs' theorem. The projection formula proves (2.5), compatibly with products and ratios.

Finally, an exceptional prime has infinite boundary multiplicity. A nonexceptional prime is a strict transform, with pullback order one and unchanged coefficient. These two cases prove (1.2). $\square$

Corollary 2.2 (Descent across exceptional centers). *In the situation of Theorem 2.1, a meromorphic m -pluricanonical form on Z' satisfying the boundary bounds of $m\Gamma'$ at every nonexceptional prime descends to a section of $m(K_Z + \Gamma)$ whenever $m\Gamma$ and $m\Gamma'$ are integral. The assertion also holds after tensoring by the pullback of a line bundle on Z .*

Proof. On the complement of the exceptional image the form is a holomorphic section of the prescribed line bundle on Z : divisorial regularity suffices on a smooth space. A local trivialization reduces extension across the remaining codimension-two analytic subset to Hartogs' extension theorem. The tensor-twisted assertion has the same proof. $\square$

Corollary 2.3 (Fibers, composition, and a fixed base). *Let $p : (Z', \Gamma') \rightarrow (Z, \Gamma)$ be a modification with the boundary rule of Theorem 2.1. Then:*

- (i) *If p is a modification over a fixed base, it induces a logarithmic modification on a very general smooth fiber. The log Kodaira dimensions of the corresponding fibers agree.*
- (ii) *A finite composition of such modifications again has boundary equal to the strict transform of the initial boundary plus the reduced exceptional divisor of the composite.*
- (iii) *If $g : (Z, \Gamma) \rightarrow T$ has smooth base and $g' = g \circ p$, then*

$$(2.6) \quad B(g', \Gamma') = B(g, \Gamma).$$

Proof. For (i), choose the parameter so that the relevant smoothness and dimension statements hold for the finitely many exceptional components and their images. A component dominating the base restricts to a reduced divisor, and its exceptional image restricts to codimension at least two. Components not dominating the base do not meet this fiber. The restricted map has the same boundary rule, so Theorem 2.1 applies on the fiber. For (ii), every prime exceptional for the composite is either newly exceptional or the transform of an earlier exceptional prime; in both cases its coefficient is one.

For (iii), fix a prime $D \subset T$. Every old prime dominating D has a strict transform, with the same pullback order and boundary multiplicity. Every new prime in the minimum defining $m(g', \Gamma'; D)$ is exceptional for p , so its boundary multiplicity is infinite. Adding these entries leaves the old minimum unchanged, including when that minimum was already infinite. $\square$

For choice independence in the singular-base definition, take a common resolution of two chosen diagrams. Their induced source boundaries coincide: old primes retain their coefficients and all primes exceptional over the initial X have coefficient one. The comparison maps satisfy Theorem 2.1, so the smooth fibrations are equivalent and their total-space and very general fiber log Kodaira dimensions agree.

2.3. Flattening relative to a fixed reference pair. To compare forms over generic points of base divisors, we need control of source divisors over higher codimension. Proper analytic flattening gives a base modification whose strict transform is flat [39, 26]. Here the strict transform is the closure of the original family over the isomorphism locus, with base torsion removed. Further base change preserves flatness. We use only the resulting equidimensionality before resolving this intermediate source.

Lemma 2.4 (Flattening and the exceptional reference). *Let $g_0 : (Z_0, \Gamma_0) \rightarrow T_0$ be a fibration from a smooth compact pair with rational SNC boundary in $[0, 1]$ to a smooth compact manifold. There are a smooth-base modification $q : T \rightarrow T_0$, a smooth-source modification $p : Z \rightarrow Z_0$, and a fibration $g : Z \rightarrow T$ with $q \circ g = g_0 \circ p$ such that, for*

$$\Gamma = p_*^{-1}\Gamma_0 + \text{Exc}(p)_{\text{red}},$$

the boundary has simple normal crossing support and every prime divisor $E \subset Z$ satisfying $\text{codim}_T g(E) \geq 2$ is p -exceptional. In particular every such prime has coefficient one in Γ , and the resulting fibration is neat with reference pair (Z_0, Γ_0) .

One may resolve prescribed proper analytic bad loci on the base during this construction. After any finite sequence of further base modifications, one can choose dominating smooth source models for which the same exceptional assertion holds with respect to the original reference Z_0 .

Proof. Flatten first, before resolving the source. After resolving the new base by further base change, denote the equidimensional strict transform by $\bar{g} : \bar{Z} \rightarrow T$, and its modification to Z_0 by $\bar{p}$. Resolve $\bar{Z}$ and all boundary data, and write

$$Z \xrightarrow{r} \bar{Z} \xrightarrow{\bar{p}} Z_0, \quad p = \bar{p} \circ r, \quad g = \bar{g} \circ r.$$

Let $N = \dim Z_0$, $b = \dim T$, and $d = N - b$. If a prime E on Z has image contained in an analytic subset $C \subset T$ of codimension at least two, equidimensionality gives

$$\dim \bar{g}^{-1}(C) \leq \dim C + d \leq N - 2.$$

It follows that $r(E)$, and therefore also $p(E)$, has dimension at most $N - 2$. Thus E is exceptional over Z_0 . The boundary and orbifold-morphism assertions follow from Theorem 2.1. The new map has connected general fiber. Its Stein factorization has a finite bimeromorphic map to the normal base T , which is an isomorphism, so all its fibers are connected.

For any further base modification $T' \rightarrow T$, pull back the flat intermediate family and take a common resolution dominating the previous smooth source. The same dimension estimate proves exceptionality over Z_0 . This includes principalizations of prescribed bad loci and their SNC resolutions on the base. Iterating proves the persistence assertion; Theorem 2.3(ii) identifies the stepwise boundary with that computed directly over Z_0 . $\square$

The resolution need not be flat: the conclusion we retain is exceptionality over Z_0 , which permits Theorem 2.2.

All constructions remain available in Fujiki class $\mathcal{C}$. A smooth compact member admits a compact Kähler modification [23, Theorems 0.7 and 3.4]. Starting with one for Z_0 , graph resolutions and principalizations by smooth-center blowups give a Kähler dominating source. Kählerness is preserved by these projective modifications, and proper meromorphic images remain in class $\mathcal{C}$ [26, Lemmas 4.4 and 4.6(3)]. Thus the bases also admit Kähler models. The preceding dimension argument still refers to the original Z_0 .

2.4. Keeping the original base fixed.

Corollary 2.5 (Simultaneous model control). *Suppose a smooth pair (X_1, Δ_1) has a fibration $f_1 : X_1 \rightarrow Y$ to a fixed smooth base, and every prime divisor contracted by f_1 into codimension at least two has coefficient one in Δ_1 . Put $C = B(f_1, \Delta_1)$. Let a factorization through another smooth compact space be given, after an allowed source modification if necessary, by*

$$X_1 \xrightarrow{g_1} W_1 \xrightarrow{h_1} Y.$$

Apply Theorem 2.4 to g_1 and make any finite number of subsequent modifications of W_1 , always taking dominating smooth source models with reduced added exceptional boundaries. The resulting diagram $X \xrightarrow{g} W \xrightarrow{h} Y$, with source boundary Δ , can be chosen to have all the following properties:

- (i) *Every prime contracted by g into codimension at least two is exceptional over the chosen smooth reference source for g_1 .*
- (ii) *Every prime contracted by $f = h \circ g$ into codimension at least two in Y has coefficient one in Δ .*
- (iii) *$B(f, \Delta) = C$. In particular, for each prime $D \subset Y$ and each prime $E \subset X$ dominating D ,*

$$(2.7) \quad \text{ord}_E(f^*D) m_\Delta(E) \geq m_C(D).$$

- (iv) *The log Kodaira dimensions of the total source and of its very general fibers over Y are unchanged by these source modifications.*

The hypothesis on contracted divisors follows from Theorem 2.4, applied to the original map to its resolved base before fixing Y .

Proof. Assertion (i) is Theorem 2.4. For (ii), a source prime is either newly exceptional, with coefficient one, or the strict transform of an old prime with unchanged image in Y . The hypothesis handles the latter case. Assertions (iii) and (iv) are Theorem 2.3; (2.7) is the defining minimum for C . Flattening the original map supplies the initial coefficient-one condition by making its contracted primes exceptional. $\square$

In applications we include the critical values and images of boundary strata among the base bad loci, so that C has SNC support. We then fix Y and use Theorem 2.5 for changes of intermediate bases.

2.5. The relative Iitaka map.

Lemma 2.6 (Relative Iitaka construction). *Let $f : (X, \Delta) \rightarrow Y$ be a fibration between smooth compact manifolds in Fujiki class $\mathcal{C}$, with Δ a rational SNC boundary in $[0, 1]$. Suppose that*

$$k = \kappa(F, K_F + \Delta_F) \geq 0$$

on very general smooth fibers. After modifications of the source, adding the reduced exceptional divisor to the strict transform of the boundary, there is a factorization

$$X \xrightarrow{g} W \xrightarrow{h} Y$$

with smooth compact intermediate space in class $\mathcal{C}$, both maps proper with connected fibers, and

$$\dim W_y = k, \quad \kappa(G, K_G + \Delta_G) = 0$$

for very general fibers of h and g , respectively. The factorization restricts to the log Iitaka fibration on very general f -fibers. These properties persist on higher models obtained by the same rule for the source boundary.

Proof. Consider all degrees clearing the denominators of Δ . Their relative adjoint direct images are coherent. First shrink to the dense analytic open where the map and boundary strata are smooth. There the family and its adjoint line bundles are flat over the base; generic constancy of the fiber section dimension permits cohomology and base change [33, §7, nos. 4–6, Sätze 3–5], with its corrigendum [34]. Together with the evaluation ranks this gives a dense analytic open for each degree; outside the resulting countable collection of exceptional sets, the maximum image dimension is k , attained in one fixed degree.

Choose a sufficiently divisible degree giving the stabilized maps, resolve the relative meromorphic map to the projective bundle of its direct image, and take Stein factorization onto the image. The Iitaka fibration theorem for line bundles on compact complex spaces identifies the resulting map on very general fibers [59, Chapter II, §5; Chapter III, §7]. No finite generation is required. The image has relative dimension k ; its Stein factor is in class $\mathcal{C}$, being a proper image of a space in that class. Resolve it and the resulting source map. Connectedness of the fibers of h follows from that of f , since their images under g are precisely the fibers of h .

The assertion that the restriction has Kodaira dimension zero concerns the original line bundle. Its standard proof applies here: coherent direct images on the projective Iitaka image acquire sections after an ample twist. If the restriction had positive Iitaka dimension, multiplying those sections by powers of the defining system would enlarge the Iitaka image. A nonzero defining section restricts nontrivially to a very general Iitaka fiber, so the restriction has dimension exactly zero.

Apply this to the resolved log adjoint on a very general f -fiber. Its divisible section ring is unchanged by Theorem 2.1, and adjunction identifies the restriction with $K_G + \Delta_G$. Further source modifications preserve the same section ring, hence the relative Iitaka factorization and its asserted fiber dimensions. $\square$

2.6. The exact comparison for Kodaira-zero fibers. For fibers of logarithmic Kodaira dimension zero, the relative pluricanonical forms determine a single Hodge line on the base. We construct that line, compute its boundary orders, and use them to identify the adjoint section spaces. Coefficient-one boundary components introduce mixed Hodge structures, but the line itself lies in one pure weight grade.

A pure variation of Hodge structures carries a local system and a varying Hodge filtration. The *highest step* of a summand is its highest nonzero filtration piece. The *parabolic extension* of such a line is obtained by local power substitutions making boundary monodromy unipotent,

extension of the unipotent Hodge filtration, and descent with the resulting rational weights. We will compute those weights using the actual relative pluricanonical form.

Proposition 2.7 (The rank-one comparison). *Let $g : (X, \Delta) \rightarrow W$ be a fibration with connected fibers from a smooth compact Kähler manifold to a smooth compact manifold in class $\mathcal{C}$. Suppose that Δ is an SNC boundary with rational coefficients in $[0, 1]$ and that $\kappa(G, K_G + \Delta_G) = 0$ on a very general smooth fiber G . After the modifications of Theorems 2.1 and 2.4, fix a smooth compact reference pair (X_0, Δ_0) and a modification*

$$\pi : X \longrightarrow X_0, \quad \Delta = \pi_*^{-1} \Delta_0 + \text{Exc}(\pi)_{\text{red}},$$

such that every prime divisor mapped by g into codimension at least two is π -exceptional. Then there are a rational line bundle M and a rational divisor T on W with the following properties.

- (i) *A positive integral multiple of M is the parabolic highest Hodge line of a complex direct summand of a pure, real-polarizable variation with an integral lattice on a dense open subset of W . Its parabolic extension is the line to which the adjoint-positivity theorem, Theorem 3.1, will apply.*
- (ii) *The divisor T is supported on an SNC divisor and satisfies*

$$(2.8) \quad B(g, \Delta) \leq T \leq 1.$$

In particular, T is effective. If every prime over a prime D of W has Δ -coefficient one, then $T_D = 1$.

- (iii) *In sufficiently divisible degrees q , multiplication by the relative generating forms gives linear isomorphisms*

$$(2.9) \quad H^0(W, q(K_W + T + M)) \simeq H^0(X, q(K_X + \Delta)) \simeq H^0(X_0, q(K_{X_0} + \Delta_0)).$$

In a common divisible grading they preserve products and ratios of sections, and hence the dimensions of the images of their linear systems.

- (iv) *The same comparison holds relatively. If $h : W \rightarrow Y$ and $f_0 : X_0 \rightarrow Y$ are holomorphic maps with $f_0 \pi = hg$, then*

$$(2.10) \quad h_* \mathcal{O}_W(q(K_W + T + M)) \simeq (f_0)_* \mathcal{O}_{X_0}(q(K_{X_0} + \Delta_0))$$

for sufficiently divisible q . The comparison of ratios also applies to systems over Y and to their restrictions wherever generic base change holds. The relative identity remains valid after tensoring by a line bundle on Y , or a rational line bundle in degrees clearing its denominator.

The Hodge construction and its divisorial calculation are independent of the reference pair. The condition on π is used in the final section comparison, to descend forms across exceptional centers.

2.6.1. The root form and its eigenspace. The cyclic-root construction and its distinguished eigen-sheaf occur in [32, Remark 2.6], [2, Lemma 5.2], and [31, Remark 3.9]. Here the order calculations also include coefficient-one boundary and keep the character line distinct from the entire top Hodge space.

Choose a positive integer m clearing the boundary denominators such that the generic rank of

$$g_* \mathcal{O}_X(m(K_{X/W} + \Delta))$$

is one. The same holds in every positive multiple of this degree: powers of a nonzero section exist, whereas two independent sections would contradict Kodaira dimension zero. Its reflexive hull $\mathcal{E}_m$ is a line bundle since W is smooth. A local frame s of $\mathcal{E}_m$ determines a meromorphic relative m -canonical form on X over the corresponding base open set. Two such frames differ by

an invertible base function. The direct image records m -pluriforms. Positivity will come from the Hodge extension of their rational root; comparing the two extension orders will supply T .

Resolve the horizontal zeros, poles, and boundary, then choose a dense open set W° on which g is smooth, the resulting divisors are relatively SNC, and all their strata are smooth over the base. By Theorem 2.1, new exceptional components have coefficient one, and the logarithmic Kodaira dimension of the general fiber stays zero. On each base chart take the normalized cyclic cover of m -th roots of s as a relative pluriform and resolve it functorially. Write τ for its tautological relative top form and $d = \dim X - \dim W$.

At a horizontal prime of boundary coefficient δ , let l be the order of s divided by m . The inequality $l \geq -\delta \geq -1$ shows that τ has at most logarithmic poles. Indeed, at a covering prime of ramification index j , its order is

$$(2.11) \quad j(1 + l) - 1.$$

This number is integral. If $l > -1$ it is nonnegative; if $l = -1$ it equals -1 . On the resolved cover let H be the reduced inverse image of the horizontal primes with $l = -1$. It includes the exceptional components above those primes that are needed in the log resolution. Logarithmic pullback preserves the bound, and no additional horizontal poles occur. We use the cohomology of the complement of H .

Lemma 2.8. *The tautological form spans the root-character part of $F^d H^d(\tilde{G} \setminus H, \mathbf{C})$ on a general fiber of the cyclic cover. In particular, that part has dimension one.*

Proof. For a compact Kähler SNC pair, logarithmic Hodge–de Rham degeneration identifies this top Hodge space with its global logarithmic top forms; it is also the smooth-fiber case of [30, Theorem 1.1], by duality. Let η be another form of the tautological character. Then η/τ is deck-invariant and descends to a meromorphic function v on G .

Consider a prime where the logarithmic section s has no zero. There $l = -\delta$. For $l > -1$, the integer in (2.11) lies between 0 and $j - 1$. A pole of v downstairs would lower the covering order by at least j , which is impossible. For $l = -1$, the logarithmic pole allowance is already exhausted, so the same conclusion holds. Thus the poles of v can occur only on the positive zero divisor of s as a section of $m(K_G + \Delta_G)$. A sufficiently high power s^N absorbs those poles. Both s^N and vs^N are then sections of $Nm(K_G + \Delta_G)$. The assumption $\kappa = 0$ forces them to be dependent, and hence forces v to be constant.

The argument also applies to a disconnected cover: we retain the whole cover with its μ_m -action, whose invariant meromorphic functions descend to G . $\square$

The ranks of the Hodge and weight graded bundles are locally constant on W° . The conclusion of Theorem 2.8, proved on a very general fiber, therefore holds throughout this smooth log locus. Notice that neither the argument nor (2.11) requires a boundary coefficient to have the form $1 - 1/a$ with a an integer.

2.6.2. Pure weights and descent of the local systems. We next place a power of the local Hodge line in a global pure variation. This requires compatible polarizations and descent, because the cyclic covers need not form a global family.

First make the local normalized-cover constructions on charts of the compact base W , using the meromorphic forms supplied by the frames of $\mathcal{E}_m$. They extend across the missing base divisor as finite covers; subsequent resolutions can be projective and functorial in the analytic category [9, Theorem 1.1]. The resulting maps to the Kähler source are projective, since the finite covers are projective. Thus their total spaces are Kähler over relatively compact source neighborhoods [26, Definition 4.1 and Lemma 4.4]. Properness allows such a neighborhood to contain the inverse image of a sufficiently small disk transverse to a base divisor. The divisor of

the tensor is unchanged under multiplication by a base unit. Thus the constructions on overlaps are identified after taking a local root of that unit, and the resolutions and their exceptional divisor classes are identified as well.

Fix a Kähler class ξ on X . On each resolved-cover chart $q_i : Z_i \rightarrow X|_{U_i}$, choose the relatively ample line bundle A_i as follows. Functoriality matches the blowup center ideals and their tautological line bundles under the root-change identifications. Insert identity stages where a center is empty; the corresponding tautological bundle is trivial there. A sufficiently weighted tensor product gives compatible A_i , equivariant under deck transformations, by descent of the centers, blowups, and tautological bundles. The classes

$$\theta_i = Nq_i^*\xi + c_1(A_i)$$

are Kähler on the smooth compact fibers for N sufficiently large; their restrictions to the compact boundary strata are Kähler too. At the intermediate finite covering stage one uses the pullback Kähler *class*; the literal pulled-back form can degenerate at ramification. The relatively ample class supplies positivity along the projective resolution. One may choose a single N : carry out the construction over charts U_i^+ containing the closures of finitely many relatively compact charts U_i covering W . Properness makes the inverse images of these closures compact, so each admits a bound N_i in this class construction. Taking $N \geq \max_i N_i$ works on all charts, since increasing N adds the semipositive class $q_i^*\xi$. This is why the construction was extended over the compact base before shrinking to W° . For the invariant Kähler-class construction on an equivariant resolved cyclic cover, see also [20, Lemma 8.1].

The classes θ_i come from total-space cohomology, so their restrictions to the smooth fibers and strata are flat for the Gauss–Manin connection. They agree under the overlap identifications. Lefschetz decomposition and cup product consequently give compatible flat real polarizations on the compact-stratum cohomology systems. The weight spectral sequence, with its Tate twists, gives the pure weight grades of the open-fiber cohomology as subquotients of these systems. They are real-polarizable: a real Hodge subvariation of a polarized pure variation is nondegenerate for the polarization, and a quotient can be identified with an orthogonal complement. These operations are compatible with the overlap maps. The integral structures come from cohomology modulo torsion and the saturated intersections with the rational weight filtration. We do not assert that the real polarizations are rational.

There is a unique weight w for which

$$F^d \operatorname{Gr}_w^W H^d(\tilde{G} \setminus H, \mathbf{C})_\chi \neq 0.$$

Here χ is the tautological character. Uniqueness follows from the rank-one assertion and strictness of the weight filtration. Let V_i be the pure integral variation given by this weight grade on the i th chart; the chosen line is $F^d(V_i)_\chi$.

Lemma 2.9. *A positive tensor power of these local Hodge lines is the highest Hodge line of a complex direct summand of a global real-polarizable integral pure variation on W° .*

Proof. Refine an overlap so that a root of its transition unit can be chosen. It gives an isomorphism between the cyclic-cover families, and therefore an isomorphism of their integral cohomology systems. Another choice differs by a deck transformation. On a triple overlap the cocycle defect is also a deck transformation. All these maps commute with the distinguished μ_m -action.

Put

$$\mathbb{U}_i = ((V_{i,\mathbf{Z}}/\text{torsion})^{\otimes m})^{\mu_m},$$

using the diagonal deck action. The ambiguity and the cocycle defect act trivially on $\mathbb{U}_i$. Hence the $\mathbb{U}_i$ descend to an integral local system $\mathbb{U}$; saturation only changes the lattice by an isogeny. The compatible real polarizations, tensorized and restricted to invariants, polarize this pure variation.

Over $\mathbf{C}$ the character projector e_χ is a flat Hodge endomorphism on each chart. The projector $e_\chi^{\otimes m}$ restricts to the invariant tensor space, because $\chi^m = 1$. It commutes with all transition maps and therefore defines a global complex direct summand with local fibers $((V_i)_\chi)^{\otimes m}$. Its highest step is $(F^d(V_i)_\chi)^{\otimes m}$, a line. In local rooted notation this line is generated by $\tau^{\otimes m}$, which is identified meromorphically with s . Thus no global root of $\mathcal{E}_m$ and no global family of cyclic covers is needed. $\square$

2.6.3. The two extension orders. Let M denote one m th of the parabolic extension of the line just constructed. We use $s^{1/m}$ for its local rational frame; the notation becomes literal after the powers and finite boundary covers above. We now compare this parabolic extension with the pole bounds of the original pluriform.

Fix a prime divisor D on W and use a local ordinary base volume ω that does not vanish at its generic point. Work on a source model on which the divisor of the generator, the boundary, and the inverse image of D have SNC support over that generic point, using strict transforms plus reduced exceptional boundary as before. All primes of this prepared source dominating D , including the exceptional primes introduced by its resolution, enter the following order tests. Put

$$l_E = \frac{1}{m} \operatorname{ord}_E(s \wedge g^* \omega^m), \quad a_E = \operatorname{ord}_E(g^* D), \quad \delta_E = \Delta_E,$$

and

$$\alpha_D = \min_{g(E)=D} \frac{l_E + \delta_E}{a_E}, \quad \beta_D = \operatorname{ord}_D^M(s^{1/m}).$$

The notation $s \wedge g^* \omega^m$ means the relative-times-base identification in the m th tensor power of the canonical bundle. For q divisible by m and clearing the rational data, and a meromorphic base function φ , the total pluriform $\varphi \omega^q s^{q/m}$ has boundary-adjusted order

$$a_E \operatorname{ord}_D(\varphi) + q(l_E + \delta_E)$$

at E . It therefore satisfies the source boundary bounds at all these primes if and only if

$$(2.12) \quad \operatorname{ord}_D(\varphi) + q\alpha_D \geq 0.$$

The Hodge frame contributes $q\beta_D$ to the order on the base, so the candidate boundary correction is $T_D = \alpha_D - \beta_D$. Computing β_D will reduce the required bounds on T_D to a comparison of two minima. We first justify computing that Hodge order by logarithmic orders on a semistable cover, without changing the chosen pure Hodge line. Global consistency of T and descent across the untested source primes will follow afterward.

2.6.4. The extension across a transverse disk. We use the analytic canonical-extension theorem [30, Theorem 1.1]. It applies to a proper surjective morphism of an analytic SNC pair (Z, H) to a smooth base, with H reduced, every stratum Kähler and dominant, and all strata smooth off a normal crossing discriminant. It supplies a graded-polarizable real variation on compact-support cohomology, locally free double Hodge/weight grades on the extension, and the dual direct-image description by $\omega_{Z/B}(H)$. Admissibility is established in [30, Remark 4.8]. These conclusions apply to proper Kähler families; projectivity is not required.

Take a transverse disk at a general point of a base divisor. Resolve the cyclic-cover pair together with the central fiber, leaving its smooth logarithmic locus over the punctured disk unchanged. Near a central point the resulting map has the form

$$t = \varepsilon x_1^{a_1} \cdots x_r^{a_r}, \quad H \subset \{x_{r+1} \cdots x_l = 0\},$$

where ε is a unit and H is the horizontal boundary. Absorbing it into a vertical coordinate gives a strict toroidal morphism. Compactness of the central fiber and properness allow us to shrink the disk until finitely many such charts cover its inverse image. The associated cone complex

is therefore finite. Apply the combinatorial semistable subdivision theorem [1, Theorem 2.7]. Its base is one ray, so the base lattice alteration is realized by $t = u^N$. Normalize this base change, keeping all components, and realize the projective source subdivision analytically [25, Lemmas 2.14–2.15, 2.18, and 2.31(4)]. Zero-image faces, which record the horizontal boundary, are allowed in the combinatorial theorem.

The semistable local form is

$$u = y_1 \cdots y_q, \quad H \subset \{y_{q+1} \cdots y_k = 0\};$$

see [25, Lemma 2.21(4)]. Here H includes every horizontal exceptional component in the transformed boundary. The total space is smooth, the central fiber is reduced, and its union with H is SNC. On every horizontal intersection stratum the displayed monomial is nonconstant; properness makes its image the whole disk. The composite to the initial Kähler total space is projective, so the new total space and its smooth strata are Kähler after shrinking the disk [26, Lemma 4.4].

These toroidal modifications preserve the open pair on the punctured fibers. Its cohomology is therefore the pullback of the original open-fiber cohomology, including the full direct sum if the cover splits. Transport the original μ_m -action through this identification. The deck projector then acts on the variation and, by functoriality, on its canonical extension; no action on a selected component or on the chosen semistable model is needed. We apply the extension theorem componentwise and take the direct sum. We use this semistable pair itself: an arbitrary further resolution need not preserve reducedness.

For application of the cited theorem, H consists of the horizontal logarithmic boundary. The central fiber is not included in H : it appears in the relative logarithmic complex, with dt/t in the base quotient. Write $f : Z \rightarrow B$ for this degeneration over the disk, and $Z_0 = f^{-1}(0)$ for its central fiber. The top relative logarithmic sheaf is

$$(2.13) \quad \omega_{Z/B}(H + (Z_0)_{\text{red}} - f^*[0]).$$

On the semistable model the central fiber is reduced, so this is $\omega_{Z/B}(H)$. For a smooth open fiber of dimension d , Poincaré duality identifies

$$H_c^d(Z_t \setminus H_t)^* \simeq H^d(Z_t \setminus H_t)(d).$$

Accordingly the dual degree-zero Hodge quotient in [30, Theorem 1.1(iv)] is precisely the top F^d step of ordinary H^d . The upper and lower canonical extensions coincide after unipotentization. This proves that the canonical extension of the top step is the direct image of (2.13). The Kähler form of the one-parameter limiting theorem is also explained in [30, Remarks 2.16, 4.4, and 4.8].

Lemma 2.10. *The projection of the rank-one top eigenspace to its nonzero pure weight grade is an isomorphism on canonical extensions after unipotentization. Consequently it is an isomorphism of parabolic rational lines before that base change.*

Proof. Let $\bar{\mathcal{V}}_\chi$ be the extended mixed eigensystem over the disk. The finite deck projector acts on the canonical extension and its filtrations. By the cited extension theorem, each

$$\text{Gr}_F^p \text{Gr}_j^W \bar{\mathcal{V}}_\chi$$

is locally free and has its rank on the punctured disk. The filtrations on the weight quotients are induced filtrations. Thus the exact sequences

$$0 \longrightarrow F^d W_{j-1} \bar{\mathcal{V}}_\chi \longrightarrow F^d W_j \bar{\mathcal{V}}_\chi \longrightarrow F^d \text{Gr}_j^W \bar{\mathcal{V}}_\chi \longrightarrow 0$$

have locally free quotients. There is no step above F^d , also on the extension, by these constant-rank assertions. Exactly one of the displayed quotients has rank one, namely that for $j = w$; all the others have rank zero. Induction on the weight filtration therefore gives

$$F^d W_{w-1} \bar{\mathcal{V}}_\chi = 0, \quad F^d W_w \bar{\mathcal{V}}_\chi = F^d \bar{\mathcal{V}}_\chi,$$

and identifies the latter line with $F^d \mathrm{Gr}_w^W \overline{\mathcal{V}}_\chi$ across the origin, without an additional zero or pole. Descent gives the parabolic assertion. $\square$

2.6.5. *The divisorial order of the Hodge line.* With the notation and prepared source above, we claim that the parabolic order of the frame $s^{1/m}$ in M is

$$(2.14) \quad \beta_D = \mathrm{ord}_D^M(s^{1/m}) = \min_{g(E)=D} \frac{l_E + 1 - a_E}{a_E}.$$

Equivalently, β_D is the largest rational number b such that $t^{-b}s^{1/m}$ extends at $D = (t = 0)$ in the parabolic sense.

It suffices to work over a transverse disk, suppressing the other base coordinates. Write $\tau = s^{1/m}$ and $\Omega = \tau \wedge g^*dt$. The multivalued total form Ω/t has order $l_E - a_E$ at E . A logarithmic extension allows order -1 , so twisting by t^{-b} imposes

$$(2.15) \quad l_E + 1 - a_E - a_E b \geq 0.$$

To verify this directly against the canonical extension, take a finite base change $t = u^N$ and a semistable model of the root cover. At a central prime over the strict transform of E , let e be the total ramification index of the composite map to X , including the cyclic-cover ramification. Reducedness gives $ea_E = \mathrm{ord}_{E'}(t) = N \mathrm{ord}_{E'}(u) = N$, where E' is this new prime. Orders of a total volume form shifted by one multiply by e . Dividing its pullback by $dt/du = Nu^{N-1}$ therefore gives

$$\mathrm{ord}(\tau \wedge du) = e(l_E + 1) - N = e(l_E + 1 - a_E).$$

The base twist subtracts $Nb = ea_E b$. Regularity in (2.13) is exactly (2.15). By Theorem 2.10, testing the logarithmic top form tests the chosen pure Hodge line as well. The form being tested is the pullback of the original tautological form on the full cover. The identification of open-fiber cohomology above and the direct-image description of its canonical extension identify it with that same Hodge section. Keeping all normalized components ensures that a prime above every original E is present; the proper toroidal modifications retain its strict transform.

To prove sufficiency, continue to write X for the smooth source over the transverse disk and put $n = d + 1$. In SNC coordinates write

$$\Omega = h \prod_i x_i^{l_i} dx_1 \wedge \cdots \wedge dx_n, \quad t = h' \prod_i x_i^{a_i},$$

where h, h' are units and $a_i = 0$ for horizontal components. Put $c_i = l_i + 1 - (1+b)a_i$. The vertical inequalities (2.15) and the horizontal logarithmic bounds give $c_i \geq 0$. For any additional divisorial valuation v , write $A_X(v)$ for its log discrepancy over this smooth model, that is, one plus the order of the Jacobian. Logarithmic pullback of the SNC coordinate divisor gives $A_X(v) \geq \sum_i v(x_i)$, exactly as in Theorem 2.1. Thus the shifted order of $t^{-(1+b)}\Omega$ is

$$A_X(v) + \sum_i (l_i - (1+b)a_i)v(x_i) \geq \sum_i c_i v(x_i) \geq 0.$$

These inequalities persist under finite covers, since shifted orders multiply by ramification indices. Further exceptional primes therefore impose no stronger condition. Horizontally, shifted orders are strictly positive outside the designated log boundary; on the root cover these are positive integers, so there are no poles there. Conversely, covering primes above strict transforms detect every inequality in (2.15). Taking their minimum proves (2.14).

Remark 2.11. The use of all primes on the resolved model in (2.14) matters even for a simple logarithmic family. On $\mathbf{P}_x^1$ over a disk, remove the horizontal divisor $x^2 = t$ and use $\tau = dx/(x^2 - t)$. After $t = u^2$ and $x = uz$, one has $\tau = u^{-1}dz/(z^2 - 1)$, so the parabolic order is $-1/2$.

Resolving the tangency of the horizontal divisor with $t = 0$ introduces a prime with $a_E = 2$ and $l_E = 0$, which gives exactly $-1/2$ in (2.14). The original central prime alone would not detect it.

2.6.6. *The boundary and the comparison of sections.* Define

$$(2.16) \quad T_D = \alpha_D - \beta_D.$$

Changing s to vs , for a meromorphic base function v , adds $a_E \operatorname{ord}_D(v)/m$ to l_E . Hence it adds $\operatorname{ord}_D(v)/m$ to both α_D and β_D and leaves T_D unchanged. This also checks the transformation rule for the rational Hodge frame in (2.14). Passing to a higher divisible generating degree has the same effect: on the generic fiber its generator is a base multiple of a power of s . Thus the construction is independent of these choices.

For the boundary inequality, write

$$x_E = \frac{l_E + \delta_E}{a_E}, \quad c_E = 1 - \frac{1 - \delta_E}{a_E}.$$

Then $0 \leq c_E \leq 1$, and

$$T_D = \min_E x_E - \min_E (x_E - c_E).$$

For any finite collections of real numbers x_E, c_E , this difference lies between $\min_E c_E$ and $\max_E c_E$. Moreover the inf-multiplicity convention gives

$$\min_E c_E = 1 - \max_{g(E)=D} \frac{1 - \delta_E}{a_E} = B(g, \Delta)_D.$$

This proves (2.8). If all $\delta_E = 1$, all $c_E = 1$, so $T_D = 1$.

Choose the discriminant to include the vertical boundary and all loci where the smooth log construction fails. Off it the family and the generator line are smooth log-relative data, and there is no vertical boundary. There $\alpha_D = \beta_D$; in a nonvanishing generator frame both are zero. Thus T has finite support in the chosen discriminant, which may be resolved to be SNC. On subsequent modifications one pulls back the variation and its parabolic line and recomputes α_D and (2.16); the order computation is unchanged on the resulting models.

Completion of the proof of Theorem 2.7. Take q divisible by m and by the denominators of all rational data. In a local base frame, a meromorphic section of $q(K_W + T + M)$ is represented by

$$\varphi \omega^q s^{q/m}.$$

The order of the frame in $T + M$ is $q(T_D + \beta_D) = q\alpha_D$. Consequently this section is regular at D exactly when (2.12) holds. The source-side order test already proved identifies this with the boundary bounds at every source prime dominating D . Along horizontal primes the required bound is automatic because $s^{q/m}$ is a logarithmic pluricanonical section on the general fiber. We have proved the exact section comparison in codimension one on the base, and at all source primes except those contracted by g into codimension at least two.

Conversely, a section of $q(K_X + \Delta)$ restricts on a general g -fiber to a multiple of $s^{q/m}$. Their ratio is a meromorphic function constant on the connected general fibers, and therefore descends meromorphically to W . The same divisorial inequalities make it a section of $q(K_W + T + M)$ outside codimension two; smoothness of W extends it uniquely across that set. This proves the converse comparison in all divisible degrees under consideration.

For the forward direction, a section on W gives a meromorphic adjoint form on X whose only possible remaining poles lie on g -contracted prime divisors. By hypothesis these divisors are π -exceptional. Push the form to X_0 . At every prime of X_0 its transform was among the tested primes of X , so the pushed form satisfies the Δ_0 pole bound in codimension one. Since X_0 is smooth and the adjoint multiple is a line bundle, it extends across codimension two. Finally Theorem 2.1 pulls it back to a section of $q(K_X + \Delta)$, proving (2.9).

If the diagram is over Y , apply this argument over an arbitrary open subset $U \subset Y$. Its inverse image in X is $\pi^{-1}(f_0^{-1}U) = g^{-1}(h^{-1}U)$, and π is still proper over the smooth open space $f_0^{-1}U$. The exceptional images still have codimension at least two. The construction and the extension of sections commute with restriction to U , giving (2.10). Relative canonical versions, when the auxiliary base is smooth, follow by the projection formula. All comparisons multiply by the same relative generator, with its powers in the common divisible grading, so they preserve products and ratios of sections. The projection formula also gives the identity after any pulled-back base twist, in degrees clearing the denominator for a rational twist. This completes Theorem 2.7. $\square$

2.7. Retaining the original base multiplicities.

Lemma 2.12 (Composition of inf-multiplicity bounds). *Suppose that $X \xrightarrow{g} W \xrightarrow{h} Y$ is a diagram of surjective maps between smooth spaces, and Δ, T, C are rational boundaries on these spaces. If*

$$T \geq B(g, \Delta), \quad C \leq B(h \circ g, \Delta),$$

then $B(h, T) \geq C$. If, in addition, every prime divisor of X mapped into codimension at least two in Y has coefficient one in Δ , and $T \leq 1$, then every prime divisor of W mapped into codimension at least two in Y has coefficient one in T .

This is the inf-multiplicity composition comparison of [16, Proposition 3.13(1)]; we include its order calculation to track coefficient-one exceptional divisors.

Proof. Let $D \subset Y$ and $V \subset W$ be prime divisors with $h(V) = D$. Write $b_V = \text{ord}_V(h^*D)$. If $E \subset X$ is a prime divisor dominating V , put $a_E = \text{ord}_E(g^*V)$. At their generic points orders multiply, so

$$\text{ord}_E((h \circ g)^*D) = b_V a_E.$$

The hypothesis on C implies $b_V a_E m_\Delta(E) \geq m_C(D)$ for every such E . Taking the minimum over them and then using the hypothesis on T gives

$$b_V m_T(V) \geq b_V m(g, \Delta; V) \geq m_C(D).$$

Taking the minimum over V proves the first assertion. All these inequalities are valid with the prescribed convention for infinity.

For the second assertion, let V be mapped into codimension at least two in Y . Every prime E dominating it has $\Delta_E = 1$, so $m(g, \Delta; V) = \infty$. It follows that $B(g, \Delta)_V = 1$, and hence $T_V = 1$. $\square$

3. THE ADJOINT-ADDITION PRINCIPLE

The section comparison reduces subadditivity to a statement about $K_W + T + M$. We prove that statement here from the two positivity inputs specified below. Its two maps have different purposes: h is the fibration whose fiber dimension is to be added, whereas p expresses the nef term as a pullback from a projective base.

The first input is the precise adjoint-positivity theorem for the line produced by the construction. We use the highest-step and parabolic-extension conventions of Section 2.6.

Theorem 3.1 (Adjoint-positivity input). *Let B be a smooth compact connected manifold in class C and $B^\circ \subset B$ a dense Zariski open set with SNC complement. Let $\mathbb{V}$ be a pure real-polarizable variation of Hodge structures on B° with an integral ambient lattice. Suppose a complex direct summand of $\mathbb{V}_{\mathbb{C}}$ has a rank-one highest nonzero Hodge step, with parabolic rational extension L . There are a modification $\tau : B' \rightarrow B$, a surjective morphism $p : B' \rightarrow S$ with connected fibers to a smooth projective variety, and a nef rational line bundle P on S such that*

$$(3.1) \quad \tau^*L = p^*P, \quad K_S + aP \text{ is big for some } a \in \mathbf{Q}_{>0}.$$

*Positive rational rescaling of L is allowed, and the line identity persists under further modifications. If S is a point, τ^*L is rationally trivial.*

The full proof is in Section 4. The lattice belongs to the ambient variation: the complex summand need not itself be defined over $\mathbf{Q}$, and the real polarization need not be rational.

The second input supplies addition when the fiber already has a big log canonical divisor. It will be proved in Section 5.

Proposition 3.2. *Let $h: V \rightarrow Y$ be a surjective holomorphic map with connected fibers between smooth compact complex manifolds in Fujiki class $\mathcal{C}$. Let T and C be effective rational divisors with simple normal crossing support and coefficients in $[0, 1]$ on V and Y , respectively. Suppose that*

- (i) $K_F + T_F$ is big for a very general smooth fiber F of h ;
- (ii) $C \leq B(h, T)$;
- (iii) $T_E = 1$ for every prime divisor E of V whose image in Y has codimension at least two.

Then

$$(3.2) \quad \kappa(V, K_V + T) \geq \dim F + \kappa(Y, K_Y + C).$$

As usual, the assertion is automatic if the last term is $-\infty$.

General-type addition will give sections after a positive ample twist from S . To remove that twist, we will need one fixed section with a negative ample twist. The next lemma produces it from a big base divisor, once nonvanishing on the general p -fiber is known. It uses weak positivity with exceptional poles and descent to a fixed smooth reference, following the method of [16, Remark 6.5(3) and Proposition 6.6]. We include the descent step explicitly.

Lemma 3.3 (Effectivity after a big base twist). *Let $p: (W, T) \rightarrow S$ be surjective, with W smooth compact in class $\mathcal{C}$, S smooth projective, and T a rational SNC boundary. Suppose one fixed divisible relative log pluricanonical system is nonzero on very general fibers. For every big rational line bundle H on S , the rational line bundle $K_{W/S} + T + p^*H$ has a nonzero section in some positive degree.*

Proof. Choose a Kähler source modification, flatten over a smooth projective modification $\pi: S' \rightarrow S$, and resolve the flat main transform. Write $\mu: W' \rightarrow W$ and $p': W' \rightarrow S'$ for the resulting maps. Give W' the strict transform of T plus the reduced μ -exceptional divisor, denoted T' . Then

$$(3.3) \quad K_{W'} + T' = \mu^*(K_W + T) + E, \quad E \geq 0 \text{ and } \mu\text{-exceptional}.$$

Every source prime over a subset of codimension at least two in S' is exceptional over the fixed W : before resolving the source, flatness bounds that inverse image by codimension at least two. Put $R = K_{S'/S} \geq 0$ and $H' = \pi^*H + R$. The exact identity

$$(3.4) \quad K_{W'/S'} + T' + p'^*H' = \mu^*(K_{W/S} + T + p^*H) + E$$

will allow descent.

Choose a divisible $m > 0$ such that $\mathcal{E} = p'_*\mathcal{O}_{W'}(m(K_{W'/S'} + T'))$ has positive rank. Generic base change and the hypothesis provide such an m . Fujino's weak positivity theorem applies to this log canonical pair and its projective base [27, Theorem 1.1]. Choose an ample Cartier divisor A_0 and an integer $\alpha > 0$ such that $H' - A_0/(m\alpha)$ is big. For some integer $\beta > 0$,

$$(\mathrm{Sym}^{\alpha\beta} \mathcal{E})^{**} \otimes \mathcal{O}_{S'}(\beta A_0)$$

is generated by global sections at the generic point.

On the locally free locus of $\mathcal{E}$, multiplication of relative sections gives an evaluation map to

$$\mathcal{O}_{W'}(m\alpha\beta(K_{W'/S'} + T') + \beta p'^* A_0).$$

It is nonzero generically, because a nonzero relative section has nonzero powers. Generic generation supplies a global section whose image is nonzero. The complement of the locally free locus has codimension at least two. Multiply a sufficiently divisible power of the evaluated section by a section of a sufficiently divisible multiple of $m\alpha\beta p'^*(H' - A_0/(m\alpha))$. The latter exists by bigness. This produces a nonzero section of a multiple of the left side of (3.4), except possibly at source primes above that codimension-two set.

Locally, a meromorphic frame of $\mathcal{E}$ expresses the evaluated section as a meromorphic combination of products of relative sections. Thus the resulting form is meromorphic, and its only possible divisorial poles lie above that omitted codimension-two set. Those primes, and E , are exceptional over W . At every prime of the original smooth W , the resulting meromorphic form therefore has the required order for a multiple of $K_{W/S} + T + p^*H$. Divisorial regularity and extension across codimension two give a global section on W . If S is a point, the same conclusion is the assumed nonvanishing. $\square$

Proposition 3.4 (Adjoint addition). *Let $h : W \rightarrow Y$ be a fibration between smooth compact manifolds in class $\mathcal{C}$. Let T, C be rational SNC boundaries with $C \leq B(h, T)$, and suppose every prime of W whose image has codimension at least two in Y has coefficient one in T . Let $p : W \rightarrow S$ be a surjective morphism with connected fibers to a smooth projective variety. Suppose P is a nef rational line bundle on S , $M = p^*P$, and $K_S + a_0P$ is big for some rational $a_0 > 0$. If $K_W + T + M$ is big on very general h -fibers, then*

$$\kappa(W, K_W + T + M) \geq \dim(W/Y) + \kappa(Y, K_Y + C).$$

Proof. The assertion is automatic if the last term is $-\infty$. Assume it is nonnegative, put $D = K_W + T$, and write $k = \dim(W/Y)$. We may first replace W by a Kähler modification, adding the reduced exceptional divisor to its strict boundary. The log section spaces, also after tensoring by the pullback of M , are unchanged. Fiberwise bigness and the inf-multiplicity boundary are preserved; newly exceptional primes have coefficient one. Thus all hypotheses remain valid.

If S is a point, M is rationally trivial and Theorem 3.2 proves the proposition. Suppose $\dim S > 0$, and choose an ample Cartier divisor A on S . A very general h -fiber is Kähler and carries the big rational line $(D + M)|_{W_y}$, so it is projective. Openness of its big cone gives a rational c with $0 < c < 1$ for which $(D + cM)|_{W_y}$ is still big. This choice works on very general fibers: choose the first fiber outside the countably many cohomology and evaluation-rank exceptional loci for rational c and divisible degrees. The one full-rank system just found then has the same rank generically. For zero-dimensional fibers this argument is unnecessary.

For every positive rational η , $cP + \eta A$ is ample. Choose a sufficiently divisible large N and a general member of the basepoint-free system $|Np^*(cP + \eta A)|$. Dividing by N gives an effective rational divisor Q_η such that

$$Q_\eta \sim_{\mathbf{Q}} cM + \eta p^*A, \quad T + Q_\eta \text{ is an SNC boundary.}$$

Bertini is applied simultaneously to the finitely many strata of T ; the member contains no old boundary component, and increasing N bounds its new coefficients by one. Existing coefficient-one components remain unchanged, and increasing the source boundary can only increase $B(h, T)$. Thus all boundary hypotheses of Theorem 3.2 hold for $T + Q_\eta$. Its fiber adjoint is big by the choice of c and nefness of the added ample pullback. Consequently

$$(3.5) \quad \kappa(W, D + cM + \eta p^*A) \geq k + \kappa(Y, K_Y + C) \quad (\eta \in \mathbf{Q}_{>0}).$$

The systems in (3.5) already have the required image dimension. It remains to remove their ample term without losing these ratios of sections. In particular one such divisor has a nonzero

section in a fixed positive degree. Restriction of that section to a very general p -fiber is nonzero; both pulled-back twists become trivial there. Hence

$$(3.6) \quad \kappa(W_s, (K_W + T)|_{W_s}) \geq 0 \quad \text{for very general } s \in S.$$

One fixed relative log pluricanonical direct image for $p : (W, T) \rightarrow S$ therefore has positive rank.

Choose rational $a > \max\{1, a_0\}$. Since P is nef, $K_S + aP$ is big. For sufficiently small rational $\lambda > 0$, $H = K_S + aP - \lambda A$ is still big. Lemma 3.3 now gives a nonzero section of a positive multiple of

$$(3.7) \quad E_- = D + aM - \lambda p^* A = K_{W/S} + T + p^* H.$$

Put

$$\theta = \frac{1-c}{a-c}, \quad \eta = \frac{(1-c)\lambda}{a-1}, \quad E_+ = D + cM + \eta p^* A.$$

Then $0 < \theta < 1$, $\eta > 0$, and direct calculation gives

$$(3.8) \quad D + M = (1 - \theta)E_+ + \theta E_-.$$

Fix a nonzero section e of rE_- for one denominator-clearing $r > 0$. For sufficiently divisible n , multiplication by $e^{n\theta/r}$ maps

$$H^0(W, n(1 - \theta)E_+) \hookrightarrow H^0(W, n(D + M)).$$

The exponent is an integer, and the source degrees range through a Veronese subsequence. On the complement of the zero divisor of e , every ratio of two sections is unchanged. These systems therefore have the same image dimensions before and after multiplication. Equation (3.5) for the displayed value of η proves the asserted lower bound. $\square$

4. ADJOINT POSITIVITY OF AN EXTREME HODGE LINE

The relative canonical bundle calculation gives a line in the highest nonzero Hodge filtration step of one complex summand. Its curvature is semipositive, but this alone does not compare it with a canonical bundle. We construct a projective quotient on which the line descends and prove that adding a positive multiple of it makes the canonical class big.

Here an *extreme Hodge line* means the rank-one highest nonzero filtration step of a complex Hodge direct summand. An integral structure is a lattice in the ambient real local system. The chosen summand need not be rational, and the ambient real polarization need not be rational either.

The *parabolic extension* is obtained by local power substitutions that make boundary monodromy unipotent, followed by Schmid extension and descent with rational weights. The monodromy theorem gives rational weights in the integral case. When the local monodromy has a finite part, these weights are essential. We first resolve any compactification whose boundary is not simple normal crossing. Unipotent extension commutes with pullback, so the same is true of the parabolic rational line after the power substitutions; see [56, 19] and [7, Lemma 5.6]. We use version 2 of [7] throughout.

We prove the adjoint-positivity theorem stated as Theorem 3.1. The line in that statement is allowed to be a complex character line inside an integral ambient variation.

There are two maps in the proof. In a flat trivialization the *line map* records only the chosen highest line; the *period map* records all Hodge filtration steps. Their ranks can differ. Curvature identifies the numerical dimension of the line with the rank of the first map. We use the full periods of suitable rational adjoint factors to construct the projective quotient. On that quotient, logarithmic general type gives a big canonical class with boundary. The remaining argument proves that the line loses numerical rank on each marked boundary component, so a section count removes the boundary after adding a multiple of the line.

4.1. Curvature, numerical dimension, and the boundary. We first relate the curvature on the open set to intersection numbers on the compactification. For a nef rational line bundle A on a compact Kähler n -fold, we use

$$\nu(A) = \max\{k : c_1(A)^k \cdot [\omega]^{n-k} > 0\},$$

where ω is any Kähler form. On a projective manifold this is the usual numerical dimension.

Lemma 4.1 (Curvature and numerical dimension). *Let T be a smooth compact Kähler manifold, let $E \subset T$ be a simple normal crossing divisor, and let A be the parabolic extension of the rank-one highest step of a polarizable pure complex variation on $T \setminus E$ with quasi-unipotent local monodromy. Then A is nef. Its numerical dimension equals the generic rank of the map that remembers this line in a flat trivialization.*

The corresponding assertion for the Griffiths line, the tensor product of the determinants of the Hodge filtration steps, uses the full period map. In particular, its rational extension is big if the full period map is generically immersive.

The curvature-and-volume argument is closely related to the proofs of [6, Lemmas 6.15 and 6.17]; here we need all mixed intersection numbers, since the chosen line need not be big.

Proof. We may check the assertions after a finite cover and resolution on which local monodromy is unipotent. This does not require an integral lattice. Indeed, $\pi_1(T \setminus E)$ is finitely generated. Put the entries of finitely many monodromy generators and their inverses in a finitely generated $\mathbf{Z}$ -algebra $R \subset \mathbf{C}$. Adjoin the finite group of roots of unity generated by the eigenvalues of the finitely many boundary monodromies, and invert every difference $\zeta - 1$ for a nonidentity element of this group. Reduction at a maximal ideal with finite residue field gives a finite-index normal congruence subgroup. Every element of a local boundary monodromy group that lies in this subgroup is unipotent: its eigenvalues belong to that finite group and reduce to one, so the inverted differences force them to equal one. This also applies to products of commuting boundary monodromies at crossings. The corresponding finite cover extends over the compactification, and a Kähler resolution gives the desired unipotent model; see [61, Lemma 2.21]. These covers preserve nefness, numerical dimension, and bigness of the rational line. Let h be its Hodge metric and Θ its curvature on $T \setminus E$, normalized to represent $c_1(A)$. Griffiths' curvature formula gives $\Theta \geq 0$; its rank is the rank of the line map, since its value on a tangent vector is the squared norm of the second fundamental form of the highest step. The same formula for the Griffiths line detects the differential of the full period map.

The Hodge norm estimates in extending frames give, in boundary coordinates $t_1, \dots, t_k$, weights whose absolute values are bounded by

$$(4.1) \quad C \left(1 + \sum_{j=1}^k \log(-\log |t_j|) \right).$$

The semipositive metric therefore extends as a positive current in $c_1(A)$ with zero Lelong numbers. Regularization of positive currents then gives nefness; see [22, Corollary 6.4]. This extends the nefness argument of [7, Lemma 2.16(1)] from a proper log smooth algebraic space to the present compact Kähler setting.

To identify numerical dimension, we must also show that no intersection mass is lost at E . Write $\alpha = c_1(A)$ and fix $\varepsilon > 0$. Since α is nef, $\alpha + \varepsilon[\omega]$ has a smooth Kähler representative θ_ε . Write

$$\Theta + \varepsilon\omega = \theta_\varepsilon + dd^c u, \quad dd^c = \frac{\sqrt{-1}}{2\pi} \partial\bar{\partial}.$$

The global potential u satisfies the local bounds (4.1). Put $n = \dim T$. If E is empty, the potential is smooth and there is no lost mass. Otherwise write $E = E_1 + \dots + E_N$ and choose

smooth divisor metrics, with defining sections s_j , such that

$$u_j = -\log |s_j|^2, \quad dd^c(-u_j) = [E_j] - \eta_j,$$

where the η_j are smooth curvature forms. Choose $c > 0$ with $c\eta_j \leq \theta_\varepsilon$ for all j , and rescale the metrics so that $cu_j \geq 1$. Then $\phi_j = -cu_j$ is θ_ε -plurisubharmonic and $\phi_j \leq -1$. For any fixed $0 < b < 1$, the attenuation theorem [36, Example 2.14] gives

$$\psi_j = -(-\phi_j)^b = -c^b u_j^b$$

with full Monge–Ampère mass in the Kähler class $[\theta_\varepsilon]$. The full-mass class is convex and contains the zero potential [36, Proposition 1.6]. Thus, for $0 < \delta \leq c^b/N$, the potential

$$(4.2) \quad v = -\delta \sum_{j=1}^N u_j^b = \sum_{j=1}^N \frac{\delta}{c^b} \psi_j + \left(1 - \frac{N\delta}{c^b}\right) 0$$

is θ_ε -plurisubharmonic and has full mass. A positive power grows faster than a logarithm, so (4.1) implies $v - C \leq u$ globally for some constant C . Full mass is preserved on passing to less singular potentials [13, Propositions 2.11(i) and 2.14(i)]. Hence u has full mass. The non-pluripolar measure gives no mass to the analytic set E and agrees with the smooth wedge product on its complement. Consequently

$$\int_{T \setminus E} (\Theta + \varepsilon \omega)^n = (\alpha + \varepsilon[\omega])^n.$$

Expand both sides as polynomials in ε . Their coefficients are finite, since all summands on the left are nonnegative. Equality for all $\varepsilon > 0$ gives

$$(4.3) \quad \int_{T \setminus E} \Theta^k \wedge \omega^{n-k} = \alpha^k \cdot [\omega]^{n-k} \quad (0 \leq k \leq n).$$

If the generic curvature rank is r , the left side is positive for $k \leq r$ and zero for $k > r$. This proves the numerical assertion. For an immersive Griffiths line the top integral is positive, and the volume criterion for a positive current on a compact Kähler manifold gives bigness; see [12]. $\square$

Lemma 4.2 (Restriction to a boundary component). *In the unipotent case of Theorem 4.1, let E_i be a smooth boundary component and N_i its monodromy logarithm. On the open stratum of E_i , the limiting highest line occurs in a unique weight grade of the limiting mixed variation. The restriction $A|_{E_i}$ is the Schmid extension, across the remaining boundary of E_i , of that graded Hodge line. Its numerical dimension is therefore the generic rank of the graded line map.*

Proof. The extended filtration and $W(N_i)$ give the limiting mixed variation along the open stratum. Since the highest step has rank one, exactly one weight grade contains it. Changes of the normal parameter act by $\exp(cN_i)$ and act trivially on the associated graded. At a further crossing the relative monodromy filtration identifies the extension of this graded variation with the graded of the original extending filtration. Thus the identification holds on the entire component. For log smooth algebraic spaces it is stated in [7, §2.5.3 and Lemma 2.16(3)]; its local proof uses the multivariable nilpotent orbit theorem [19] and applies here as well. The filtrations and strictness also restrict to a complex summand. Apply Theorem 4.1 on E_i . $\square$

4.2. Rational adjoint periods and descent of the line. The period-quotient construction requires discrete monodromy. A projection to one complex factor need not retain it. We therefore construct a rationally polarized adjoint variation from the ambient integral local system and keep whole rational factors. A tensor construction then recovers a positive power of the original line, including its scalar monodromy.

Lemma 4.3 (Rational adjoint reduction). *In the setting of Theorem 3.1, let G be the identity component of the rational Zariski closure of monodromy. Then G is semisimple. Its adjoint local system is a rationally polarized integral variation of weight zero.*

There is a monodromy-invariant collection of rational simple adjoint factors, each containing a complex factor acting nontrivially on the highest-line orbit, with the following properties. The resulting adjoint variation $\mathbb{A}$ has discrete integral monodromy. After a positive power and up to a constant one-dimensional complex Hodge shift, the original line is the highest line of a complex subvariation of a tensor construction on $\mathbb{A}$. On a finite cover preserving the factors, that line splits into the tensor product of the highest lines belonging to the individual complex factors. All these highest steps have rank one.

Proof. Semisimplicity and the theorem of the fixed part hold for polarizable real and complex variations on a Zariski open subset of a compact Kähler manifold. We may use a Kähler modification of B to apply them; see [54, Lemma 2.3 and its proof] and [5, Theorem 3 and Remark 4]. Thus the algebraic monodromy group is reductive. Pass temporarily to a finite cover on which it is connected. The irreducible complex constituents of the local system carry polarizable complex variations. They can be defined over a number field: the representation is rational and its algebraic monodromy is reductive. The determinant of each constituent is unitary, as are all its algebraic conjugates, which are again constituents of the rational representation. Its monodromy values are algebraic units by the integral lattice. Kronecker's theorem makes these values roots of unity in a fixed number field, hence makes the determinant character finite. It is consequently trivial on G . This is Deligne's finite-determinant argument; its classical algebraic formulation is [21, Corollary 4.2.8(iii)(b)]. The connected center of G acts by scalar characters on the irreducible constituents; their determinants detect these characters. Since the original representation is faithful, this connected center is trivial. Therefore G is semisimple.

We next equip the monodromy Lie algebra with a rational polarization. In every tensor construction, the subspace of G -invariant tensors is the fixed part and has a constant Hodge structure. In a flat trivialization the Hodge circles h_x and h_{x_0} therefore induce exactly the same linear operator on each such subspace at each parameter $z \in S^1$. Reductivity identifies G with the pointwise stabilizer of all its invariant tensors. The circles preserve these subspaces, so normalize G , and the equality of their actions gives

$$h_x(z)h_{x_0}(z)^{-1} \in G(\mathbf{R}).$$

Here is a direct local conjugacy argument. Along a smooth path in the base, write $a_t = \text{Ad } o h_t : S^1 \rightarrow G^{\text{ad}}(\mathbf{R})$ and $u_t(z) = (\partial_t a_t(z))a_t(z)^{-1}$. Differentiating the homomorphism identity gives

$$u_t(z z') = u_t(z) + \text{Ad}(a_t(z))u_t(z').$$

Normalized Haar averaging, with $X_t = \int_{S^1} u_t(z') dz'$, yields $u_t(z) = X_t - \text{Ad}(a_t(z))X_t$. The solution of $g'_t g_t^{-1} = X_t$, $g_0 = 1$, therefore gives $a_t = g_t a_0 g_t^{-1}$. Lift this path through the central isogeny $G(\mathbf{R})^+ \rightarrow G^{\text{ad}}(\mathbf{R})^+$. The ratio of h_t and $g_t h_0 g_t^{-1}$ lies in $G(\mathbf{R})$ by the invariant-tensor equality, and centralizes G by the equality of their adjoint actions. It lies in the finite center of G ; continuity in the circle parameter makes it the identity. Thus $h_t = g_t h_0 g_t^{-1}$, proving that the lifted period map lies in a single $G(\mathbf{R})^+$ -orbit.

It follows that $\mathfrak{g}_{\mathbf{Q}} = \text{Lie}(G)$ is a Hodge sub-local system of $\text{End}(\mathbb{V}_{\mathbf{Q}})$, with its weight-zero grading. The Weil involution θ preserves $\mathfrak{g}_{\mathbf{R}}$ and is the restriction of the Cartan involution of the real polarization isometry group. To check the restriction, use the positive Hodge metric: on its Lie algebra $\theta(X) = -X^*$. If $\mathfrak{g}_{\mathbf{R}} = \mathfrak{k} \oplus \mathfrak{p}$ is its eigenspace decomposition, then $\mathfrak{k} \oplus i\mathfrak{p}$ is a compact real form of $\mathfrak{g}_{\mathbf{C}}$, represented by skew-Hermitian endomorphisms. Thus $-B_{\mathfrak{g}}(X, \theta X)$ is positive definite, proving that θ is Cartan on $\mathfrak{g}_{\mathbf{R}}$. The appropriately signed Killing form therefore

polarizes this weight-zero variation. The form is rational. Moreover

$$\mathfrak{g}_Q \cap \text{End}(\mathbb{V}_Z)$$

is a monodromy-invariant lattice. This gives the rationally polarized integral adjoint variation.

Let ℓ be the highest line at a reference point of the chosen complex summand. The non-lowering parabolic of the Hodge grading preserves ℓ , so a Borel subgroup preserves it. Its closed orbit

$$J = G_{\mathbb{C}}\ell$$

is a flag variety, a product of flag varieties for the complex simple factors. The span W of this orbit is irreducible. Indeed, in a semisimple decomposition of the representation, the nonzero components of a Borel-eigenline all have the same highest weight; their span is a single diagonal copy of that irreducible representation. The line is its unique highest Hodge step. Indeed, at every point the Hodge circle normalizes G and preserves the highest line there; it consequently preserves the span of its $G_{\mathbb{C}}$ -orbit. Thus W is a Hodge subvariation. The lifted line map takes values in J , and the original line is the pullback of the tautological line there, equivariantly for monodromy.

Call a complex simple factor *visible* if it acts nontrivially on J . Retain every rational simple adjoint factor containing a visible factor. This collection is invariant under the possibly disconnected monodromy group. On its Lie algebra we obtain the variation $\mathbb{A}$; its monodromy is contained in the automorphisms of a lattice and is discrete. Notice that an entire rational factor is kept. Projection to only one real factor would not ensure discreteness.

We now recover the line from this adjoint system. Choose a positive integer N for which the highest weight $N\lambda$ of W belongs to the root lattice. The Cartan component $W_N \subset W^{\otimes N}$ is generated by the orbit of $\ell^{\otimes N}$. It factors through the connected adjoint group, and its highest Hodge line is $\ell^{\otimes N}$. The same orbit-span argument just used shows that W_N is a Hodge subvariation.

We must also account for disconnected monodromy before applying tensor generation. Deck transformations take the lifted line to another point of the same connected orbit, so they preserve W and its Cartan components. The kernel of the action on the retained adjoint factors centralizes the G -action on W , and hence acts scalarly by irreducibility. Its connected part is semisimple and has no nontrivial scalar character. Its component group is finite. Increasing N therefore kills the entire kernel action on W_N . Consequently W_N is a representation of the full projected algebraic monodromy group, not just of its identity component. This step rules out a discarded nontorsion scalar local system.

The adjoint representation of that projected group is faithful and self-dual. By tensor generation and complete reducibility, W_N embeds as a flat direct summand in a tensor construction $\mathbb{T}(\mathbb{A})_{\mathbb{C}}$; see [48, Theorem 4.14]. To make the embedding compatible with Hodge structures, apply the theorem of the fixed part to

$$H^0(B^\circ, \text{Hom}(W_N, \mathbb{T}(\mathbb{A})_{\mathbb{C}})).$$

This nonzero space of flat maps has a constant complex Hodge structure. A nonzero Hodge-homogeneous component of any nonzero flat map is still flat. Its kernel is G -invariant, so the map is injective because W_N is irreducible. A constant one-dimensional complex Hodge shift makes it a morphism of variations; that shift changes neither the underlying holomorphic line nor its boundary extension. This is the constituent and fixed-part argument of [5, Theorem 3].

The complex summand carries a flat Hermitian polarization. In a real tensor variation of weight w polarized by Q , it is induced by $i^w Q_{\mathbb{C}}(v, \bar{u})$, up to the conventional overall sign. Its restriction to a Hodge subvariation is nondegenerate because it is definite with the prescribed sign on each Hodge component. There is no need for the real bilinear form itself to restrict nondegenerately to the chosen complex summand: it may pair that summand with its conjugate.

Finally, on a factor-preserving finite cover an irreducible representation is a tensor product over the simple factors. The dimension of its highest step is the product of the corresponding dimensions. Since that dimension is one, each factor has a unique rank-one highest step. This gives the asserted factor lines. Their extensions are nef and functorial by Theorems 4.1 and 4.2. $\square$

Lemma 4.4 (Generic stabilizers of adjoint periods). *Let G be a connected semisimple adjoint group over $\mathbf{Q}$, and let $\mathbb{A}_{\mathbf{Q}}$ be a rationally polarized integral variation whose fiber is $\mathfrak{g}_{\mathbf{Q}} = \mathrm{Lie}(G)$ and whose connected algebraic monodromy is G , acting by its adjoint representation. No non-identity rational Lie algebra automorphism of $\mathfrak{g}_{\mathbf{Q}}$ fixes all generic lifted periods of $\mathbb{A}$. The automorphism need not belong to the monodromy group.*

Proof. Suppose γ fixes those periods. By equivariance, every monodromy conjugate of γ is everywhere of Hodge type $(0,0)$ in the endomorphism variation. Their rational span is therefore a rational subvariation of type $(0,0)$. Its induced polarization is positive definite. Intersection with the endomorphism lattice is a full monodromy-invariant lattice: clear one denominator of γ ; integral monodromy conjugation preserves that denominator. A positive definite integral monodromy group is finite. Consequently the connected group G fixes γ , so γ centralizes every inner automorphism of $\mathfrak{g}$. The centralizer of the inner group in $\mathrm{Aut}(\mathfrak{g})$ is trivial: on each simple ideal commutation with all adjoint operators makes an endomorphism scalar, and compatibility with the Lie bracket makes this scalar one; permutations of distinct ideals do not commute with their separate inner actions. Thus $\gamma = 1$. $\square$

4.3. A projective quotient for the retained periods. We now quotient by connected fibers of the retained full period map. The preceding reduction supplies both discrete integral monodromy and a tensor description of a power of the line. The first property allows us to compactify the quotient; the second will descend the line with its exact parabolic extension. No bigness assertion about $K_S + aP$ is used in this construction.

Proposition 4.5 (The projective period quotient). *In Theorem 3.1, after modifications there is a map $p : B' \rightarrow S$ with connected fibers to a smooth projective variety and a nef rational line P with $\tau^*L = p^*P$. On a dense open subset S° , the retained adjoint variation of Theorem 4.3 descends, has the same connected algebraic monodromy, and has a generically immersive period map. The line P is its parabolically extended highest-weight line, up to the positive scaling already described. If the retained collection is empty, L is rationally trivial and S may be a point.*

The construction uses the compact-period-quotient approach of [6, Theorems 7.6–7.7], whose compactification input goes back to [57, Proposition III and Remark III-C]. We use the precise class- $\mathcal{C}$ formulation of [61, Theorem 2.19] below.

Proof. Choose a compact Kähler modification of the class- $\mathcal{C}$ source. Then pass to a finite level at which the projected monodromy is neat and preserves the factors. Finite covers of the open locus extend to finite ramified covers of compactifications and then to Kähler resolutions; see [61, Lemma 2.21]. Resolve the boundary to simple normal crossings. Extend the period map over all finite-local-monodromy strata. At neat level their monodromy is trivial and the nilpotent orbit theorem gives this extension.

The extended period map on this locus is proper; see [35, Proposition 9.11 and the proof of Theorem 9.6]. Indeed, if a sequence tending to the boundary has period images in a compact subset of the quotient, translate period lifts into a compact set. The punctured-disk Schwarz estimate makes the displacement of each vanishing-coordinate monodromy tend to zero. The discrete group acts properly, since its isotropy in the real period domain is compact. The local monodromies must therefore be trivial at neat level, so such a sequence cannot escape the extended locus.

Take the Stein factorization of this proper map. Its first map has compact connected fibers and its base is finite over the period image. It extends after modifications to a map of compact class- $\mathcal{C}$ spaces with Kähler resolutions by [61, Theorem 2.19] (version 1). That theorem applies to proper connected-fiber maps from Zariski open subsets of compact class- $\mathcal{C}$ spaces. In terms of cycles, one first flattens the proper map, sends its irreducible general fibers to a component of the Douady space of the compact source, and takes the closure of that family. The component is compact and in class $\mathcal{C}$. Near a general fiber the map to the cycle space is an open embedding, since an irreducible compact cycle of the same dimension in a sufficiently small neighborhood maps to a compact analytic subset of a small Stein neighborhood in the base, hence to a point, and must be the whole irreducible fiber. The incidence map is consequently a modification of the source. This produces the required compact quotient and also shows its uniqueness as the quotient by the connected general period fibers.

On a dense open subset of this quotient the period and its integral variation descend. At neat level a constant period has trivial stabilizer; local slices to the connected-fiber map give the descent data. After shrinking to a proper smooth fibration, the map on fundamental groups is surjective, so the descended variation has the same connected algebraic monodromy. Its period differential is generically injective by construction. On a Kähler compactification its Griffiths line is big by Theorem 4.1; equivalently, one can apply [14, Corollary 1.3]. The quotient therefore has a smooth projective model. This is also the argument in [61, proof of Corollary 2.22].

The finite deck group acts on the connected-period-fiber quotient. Take its finite quotient and resolve. This again has a smooth projective model. A general fiber before this finite quotient is the union of connected level fibers indexed by one deck-group orbit. The group acts transitively on these fibers, so the quotient is the continuous image of any one of them and is connected. Thus the induced map from the original source has connected general fibers, hence connected fibers after its Stein factorization. The latter does not change its function field. All maps can be resolved using modifications of the original source and of this projective model.

There is an actual adjoint variation on a dense Zariski open subset of the quotient before changing level. Write Γ' for the normal level subgroup of Γ . The induced action of Γ/Γ' on the level quotient is faithful. Indeed, if $\delta \in \Gamma$ acts trivially there, discreteness shows locally that its action on lifted periods agrees with a fixed $\eta \in \Gamma'$. Then $\eta^{-1}\delta$ fixes a period germ, hence the whole generic lifted period image by analytic continuation. By Theorem 4.4 it is the identity, so $\delta \in \Gamma'$. Resolve the finite group action equivariantly and remove its proper algebraic fixed loci and the branch locus. On the resulting dense Zariski open subset the level quotient is free and unramified. Its equivariant integral adjoint local system and Hodge filtration descend there. In particular, this descent uses an open subset, not merely very general points.

The highest-weight construction in Theorem 4.3 gives a line on that open subset whose pullback is a fixed positive power of L . Take its parabolic extension on a log smooth projective compactification S and divide by that power to obtain P . Nefness follows from Theorem 4.1. Pullback of the unipotent extensions, followed by rational descent, gives $\tau^*L = p^*P$ on the compact models, including the boundary. Equivalently, this can be checked at level and then descended by the norm: a line whose finite pullback is trivial is torsion in the rational Picard group. The identity therefore includes the boundary and persists under further modifications.

If no factor is retained, the orbit is a point and the only possible scalar monodromy is the finite scalar group treated in Theorem 4.3. Its parabolic line is rationally trivial, as asserted. $\square$

4.4. Loss of numerical rank on the marked boundary. Fix the projective quotient of Theorem 4.5 and a log smooth compactification (S, E) of its descended variation. Let $D \leq E$ be the reduced sum of components with nonidentity local monodromy on the retained adjoint

system, including finite monodromy. Logarithmic general type will give bigness of $K_S + D$. The next lemma is what permits us to remove D .

Lemma 4.6 (Strict loss of rank at the marked boundary). *For every component D_i of D ,*

$$(4.4) \quad \nu(P|_{D_i}) < \nu(P).$$

Proof. Write $n = \dim S$ and $r = \nu(P)$, the generic rank of the line map j by Theorem 4.1. The full retained period map has rank n ; the argument does not require $r = n$. Work first on a finite Galois cover of the open locus with connected algebraic monodromy and no factor permutations. Its connected adjoint group is denoted by G .

The Ax–Schanuel decomposition. Choose a Hodge-generic point, also general for the ranks of the factor period maps and their joint differentials. There are only finitely many rational normal-factor decompositions of G , so these requirements can be imposed simultaneously. Let T be a local fiber germ of j there, and Z its irreducible Zariski closure in the source of the variation. Thus $\dim T = n - r$. Move within the germ to a smooth point of Z and use a smooth open subset of a resolution when applying Hodge theory.

The restricted variation on Z has the same generic Mumford–Tate group as the ambient variation, since Z contains the chosen Hodge-generic point. The monodromy normality theorem therefore makes its connected monodromy H normal in G ; see [4, §5, Theorem 1]. In adjoint notation write

$$G = H \times Q.$$

The Q -period is constant on the lifted Z , by the fixed-part theorem after killing finite monodromy. The varying monodromy domain on Z is therefore the domain for H . Write J_H for its line-orbit projection and $d_H = \dim J_H$. In local flat coordinates write $j = (j_H, j_Q)$, with targets J_H and J_Q , the products of line orbits for the two groups. We will show that the H -line map has full rank d_H on Z , whereas the Q -line map and the full Q -period map have equal generic rank on the ambient base.

For the first assertion, put $m = \dim Z$ and $d = \dim \check{D}_H$, where $\check{D}_H$ is the compact dual of the connected-monodromy period domain. The projection $\check{D}_H \rightarrow J_H$ is a surjective morphism of homogeneous projective flag varieties. All its fibers have dimension $d - d_H$. Let $V \subset Z \times \check{D}_H$ be the algebraic locus where the projected line equals its value on T . It follows that $\dim V = m + d - d_H$. The period graph has dimension m in an ambient variety of dimension $m + d$, so its expected intersection dimension with V is $m - d_H$.

By [8, Theorem 1.1], an intersection component of larger dimension projects into a proper algebraic weak Mumford–Tate subvariety of Z . This is impossible for the component containing the lifted period graph over T , since T is Zariski dense in Z . Conversely, a map to J_H has fibers of dimension at least $m - d_H$. At the chosen smooth point the restricted fiber is exactly T , because the ambient local fiber is $T \subset Z$. Equality follows, proving the claim. The cited theorem applies to the rationally polarized integral adjoint variation on a smooth algebraic base and uses precisely its connected-monodromy domain.

Let q denote the full Q -period map and set $s_Q = \text{rank}(dq)$. Choose the smooth point $x \in T \cap Z_{\text{reg}}$ inside the ambient constant-rank neighborhood fixed above. Since T is Zariski dense in Z , such a point exists. The constant-rank theorem gives $\ker(dj_x) = T_x T \subset T_x Z$, and hence

$$\text{rank}(dj_x|_{T_x Z}) = \dim Z - \dim T = d_H.$$

The Q -period is constant on Z , so $T_x Z \subseteq \ker(dq_x)$. Thus, at this same point,

$$\ker(dj) \subseteq \ker(dq), \quad dj_H(\ker(dq_x)) = T_{j_H(x)} J_H.$$

The first inclusion uses $T_x T \subset T_x Z$; the second is the just-established rank calculation on $T_x Z$. The differential dj_Q factors through dq . If $dj_Q(v) = 0$, choose $w \in \ker(dq)$ with $dj_H(w) = dj_H(v)$.

Then $v - w \in \ker(dj)$, and therefore $dq(v) = 0$. We conclude

$$(4.5) \quad \text{rank}(dj_Q) = s_Q, \quad r = d_H + s_Q.$$

These are generic identities by our choice of the point.

The factor-line construction, after a common positive rescaling and a further finite cover, now gives nef lines with

$$(4.6) \quad \pi^*P = P_H + P_Q, \quad \nu(P_H) = d_H, \quad P_Q = q^*B_Q.$$

Here $q : \tilde{S} \rightarrow S_Q$ is a resolved projective quotient for the full Q -period, constructed by Theorem 4.5, and $\dim S_Q = s_Q$. The highest-line rank on S_Q is s_Q by (4.5). Only the quotient construction, not adjoint positivity, has been used to construct S_Q .

Reduction for a boundary component. For a fixed marked D_i , use a common finite Galois cover that preserves the factors and makes boundary monodromy unipotent. Resolve it and the map q equivariantly, and let E_i be the strict transform of a component above D_i that is generically finite over D_i . The inertia group at its generic point fixes E_i pointwise. Further rescaling does not change any numerical dimension. Nefness gives

$$\nu(P_H|_{E_i}) \leq d_H.$$

For example, every intersection involving E_i is bounded by the corresponding intersection with a sufficiently ample effective divisor; the vanishing of higher powers of P_H then gives this inequality. If E_i does not dominate S_Q ,

$$\nu(P_Q|_{E_i}) \leq s_Q - 1,$$

and the binomial expansion for the two nef lines in (4.6) proves the strict drop. We may therefore suppose E_i dominates S_Q . At its generic point the Q -variation is pulled back from the interior of S_Q , so its local unipotent logarithm vanishes.

Infinite local monodromy. If the original local monodromy is infinite, its logarithm at unipotent level is a nonzero rational element $N \in \mathfrak{h}$. Every nonzero rational element of a rational simple factor has nonzero projection to every conjugate complex factor. Each retained rational factor has a visible complex factor. Hence N acts nontrivially on J_H .

Let w be the pure weight of the representation supplying P_H , and let $F^a = \mathbf{C}v$ be its limiting highest line at a general point of E_i . This line has a unique weight contribution, say of weight $w + k$. It is primitive and $k \geq 0$: a nonprimitive contribution, or a contribution below the middle weight, would come by monodromy Lefschetz from a nonzero filtration piece above F^a . Moreover

$$(4.7) \quad N^{k+1}v = 0, \quad N^k v \neq 0.$$

To see that this statement holds for the actual vector rather than only its weight-graded image, use the Deligne splitting of the limiting mixed Hodge structure. The one-dimensional highest step occupies a single summand, and N has degree $(-1, -1)$ for this functorial splitting. Primitive vanishing on the graded piece therefore gives the first equality in (4.7); the Lefschetz isomorphism gives the second. For a complex summand the same argument is made in its conjugate-paired real variation and restricted to the summand.

In nilpotent-orbit coordinates $[v] \in J_H$. Since this orbit is closed in projective space and invariant under $\exp(tN)$,

$$(4.8) \quad [N^k v] = \lim_{t \rightarrow \infty} \exp(tN)[v] \in J_H \cap \mathbb{P}(\ker N).$$

This fixed locus is a proper closed subset of the irreducible variety J_H , because N acts nontrivially on it. Its dimension is less than d_H . Restrict to a simply connected open part of the boundary

stratum on which k is constant. In a flat trivialization, both N and $W(N)$ are constant. Project $N^k v$ to $\mathrm{Gr}_{w-k}^{W(N)}$ and apply the inverse of the fixed Lefschetz isomorphism

$$N^k : \mathrm{Gr}_{w+k}^{W(N)} \longrightarrow \mathrm{Gr}_{w-k}^{W(N)}.$$

This recovers the weight-graded limiting highest line. Every actual vector $N^k v$ lies in $W(N)_{w-k}$, and its image in that graded piece is nonzero. Thus the graded line map factors through the rational map on

$$J_H \cap \mathbb{P}(\ker N) \cap \mathbb{P}(W(N)_{w-k})$$

induced by this fixed linear quotient and inverse. This locus is contained in the proper fixed locus in (4.8), and the rational map is defined at every point under consideration. Its rank is less than d_H . By Theorem 4.2,

$$\nu(P_H|_{E_i}) < d_H.$$

Together with $\nu(P_Q|_{E_i}) \leq s_Q$ this proves the required inequality in the infinite-monodromy case.

Finite nonidentity local monodromy. Let γ be the original finite local monodromy. The disconnected-monodromy argument in Theorem 4.3 shows that γ preserves the orbit span W . It therefore acts on the highest-line orbit J , permuting its nontrivial simple-factor flag varieties. Equivariance preserves the kernel of $G_{\mathbf{C}} \rightarrow \mathrm{Aut}(J)$, which is the product of the invisible simple factors. Thus visible and invisible factors cannot be exchanged. At unipotent level its logarithm is zero, so the variation extends with pure limiting flags. The inertia generator fixes the generic boundary point; equivariance therefore makes these flags fixed by γ . Suppose that the boundary line retained rank $r = d_H + s_Q$. Since E_i dominates S_Q , its full Q -period and its Q -line map both have rank s_Q . Along a generic fiber of this Q -period, the H -line map must consequently have rank d_H . Its coordinates fill an open subset of J_H while the full Q -flags remain fixed.

The fixed-point condition for γ rules out permutations involving any visible H -factor, including a permutation with a factor whose coordinate is held fixed. On each visible H -factor of the highest-line orbit it forces the identity action. The action of a simple adjoint group on a nontrivial flag variety is faithful; equivariance then makes the induced automorphism of that simple group the identity as well. Rationality forces the same conclusion on all conjugate factors. Thus γ acts identically on H and preserves Q .

Its Q -action fixes every generic Q -period, by the domination of S_Q and analytic continuation from the resulting open set of lifted periods. Apply Theorem 4.4 to the descended Q -variation. This action is the identity. Hence γ is the identity on the whole retained adjoint system, contradicting the marking.

In both cases nef binomial expansion and (4.5) give $\nu(\pi^* P|_{E_i}) < r$. Numerical dimension of a nef line is unchanged by a generically finite pullback, so this is (4.4). If $r = 0$, the local line fiber is the whole source germ, so its Zariski closure is the whole source and $H = G$, $Q = 1$. The Ax–Schanuel calculation gives $d_H = 0$. Every retained rational factor contains a factor acting nontrivially on the line orbit; hence there can be no retained factor. Generic immersivity of the full retained period map then forces S to be a point, with no marked component. $\square$

4.5. Removing a boundary by counting sections. The period argument must pass from a big logarithmic canonical divisor to a big adjoint divisor without its boundary. The following lemma isolates the last step: sections lost on the boundary grow more slowly in the nef direction than sections on the whole space. For a nef rational divisor J on a smooth projective n -fold, set

$$\nu(J) = \max\{q \in \{0, \dots, n\} : J^q \cdot H^{n-q} > 0\},$$

where H is ample. This is independent of H and equals zero on a point. The proof below fixes the nef coefficient before taking the asymptotic section limit.

Lemma 4.7. *Let Z be a smooth projective variety, J a nef rational divisor, A a rational divisor, and $D = \sum_{i=1}^s d_i E_i$ an effective rational divisor with smooth prime components. Suppose $A + D$ is big and*

$$\nu(J|_{E_i}) < r := \nu(J) \quad \text{for every component of } D.$$

Then $A + tJ$ is big for every sufficiently large rational t .

Proof. If $r = 0$, the condition forces $D = 0$, so A is already big; adding a nef divisor preserves bigness. This also covers dimension zero with the usual convention. Assume $r > 0$ and put $n = \dim Z$. Choose an ample rational divisor H_1 with $A + D - H_1$ rationally effective. For fixed rational $t \geq 0$ and sufficiently divisible m ,

$$h^0(Z, m(A + D + tJ)) \geq h^0(Z, m(H_1 + tJ)).$$

Since $H_1 + tJ$ is ample, asymptotic Riemann–Roch makes the leading coefficient on the right $(H_1 + tJ)^n/n!$. As a polynomial in t , this intersection has degree r and positive leading coefficient.

Remove mD one component at a time by divisor exact sequences. Every lost quotient is a line bundle on an E_i of the form

$$\mathcal{O}_{E_i} \left(m(A + D + tJ) - \sum_h k_h E_h \right), \quad 0 \leq k_h \leq md_h.$$

Choose an ample integral divisor B_0 so large that a fixed denominator-clearing multiple of $B_0 - A - D$ is globally generated. For each h , choose an ample integral divisor B_h such that both B_h and $B_h + E_h$ are globally generated. Put $H_2 = B_0 + \sum_h d_h B_h$. The difference between $m(H_2 + tJ)$ and the displayed quotient divisor is

$$m(B_0 - A - D) + \sum_h ((md_h - k_h)B_h + k_h(B_h + E_h)).$$

It is globally generated in these divisible degrees. A section not vanishing identically on E_i gives an injection into $\mathcal{O}_{E_i}(m(H_2 + tJ)|_{E_i})$. There are exactly md_i quotient terms on E_i . Hence

$$h^0(Z, m(A + tJ)) \geq h^0(Z, m(H_1 + tJ)) - m \sum_i d_i h^0(E_i, m(H_2 + tJ)|_{E_i}).$$

For each fixed t , divide by $m^n/n!$ and let m tend to infinity through the specified multiples. All the comparison divisors are ample, so the resulting lower bound is

$$(H_1 + tJ)^n - n \sum_i d_i (H_2 + tJ)^{n-1} \cdot E_i.$$

Each restriction polynomial has degree at most $\nu(J|_{E_i}) \leq r - 1$, because the restricted divisor is nef. The positive degree- r term of the first polynomial therefore dominates the loss for every sufficiently large t . The section growth of $A + tJ$ is then positive in dimension n , which is exactly bigness. $\square$

Taking $A = K_Z$ and D reduced gives the required removal of the period boundary once its restriction ranks have been established. The weights also allow simultaneous removal of other effective divisors satisfying the same rank inequality, for example a divisorial ramification term on a generically finite cover.

Proof of Theorem 3.1. Apply Theorem 4.5. If its target is a point there is nothing further to prove. Otherwise mark the divisor D as above. The adjoint variation extends across every unmarked boundary component, including their intersections away from D , because its local monodromies there are trivial. Its period differential is generically injective. Consequently [14, Theorem 1.1] gives

$$K_S + D \text{ big.}$$

If an initial choice omitted codimension-two loci, apply the theorem on a log resolution and push the sections down; the resulting logarithmic canonical divisor pushes forward to $K_S + D$. By Theorem 4.6, each marked component has strictly smaller numerical rank. Apply Theorem 4.7 with $A = K_S$ and the marked reduced divisor D . It gives $K_S + aP$ big. Together with the rational line identity in Theorem 4.5, this proves (3.1). Since P is nef, the same bigness holds for every larger rational coefficient a . $\square$

5. ADDITION WITH A LOG-GENERAL-TYPE FIBER

We prove subadditivity when the fiber is of log general type. The base may be nonprojective: the projective positivity theorem will be applied to a stable family over the image of its classifying map.

We prove Theorem 3.2.

The proof constructs finitely many relative pluricanonical forms whose ratios give a generically finite map on a general fiber. Their poles must be controlled at every vertical divisor, including divisors whose image on the original base has codimension at least two. Multiplying these forms by base pluricanonical sections then retains both the fiber coordinates and the base Iitaka coordinates.

We first construct a projective stable family over the image of a classifying map. Positivity is applied to that family. Its sections are pulled back, descended through a finite cover, and tested at vertical valuations. All degrees below clear the denominators of the relevant divisors.

5.1. Comparison with a stable family. We use the stable-family convention of [46, Notation 3.7, Remark 3.8, and Definitions 3.9 and 3.11]. A stable pair has semi-log-canonical singularities and an ample rational log canonical divisor. A stable family over a normal base is flat, proper and surjective with connected stable fibers; its boundary support avoids the generic points and the codimension-one singular points of every fiber. Its relative log canonical divisor is rational Cartier and relatively ample, so the family is projective. Boundaries are pulled back by divisorial restriction, and Cartier multiples of the relative log canonical divisor commute with base change between normal bases. Semi-log-canonicity allows nonnormal fibers and tests log canonicity on the normalization with the conductor boundary. The general fibers in our application are additionally Kawamata log terminal (klt): they are normal and all their log discrepancies are positive. This extra condition will be checked before applying positivity. A stable family has *maximal variation* when its classifying map to the moduli space of stable pairs is generically finite onto its image.

Fiberwise finite generation gives an embedding degree separately on each fiber. The next lemma selects one degree on an analytic base. As throughout, *very general* excludes a countable union of proper closed analytic subsets.

Lemma 5.1. *Let Y be an irreducible compact complex space. For each positive integer j , suppose that $q_j: Q_j \rightarrow Y$ is a proper surjective map from an irreducible compact complex space and that, over a dense analytic open $U_j \subset Y$, its fibers are irreducible of a fixed dimension r_j . Let*

$$\phi_j: Q_j \dashrightarrow Z_j$$

be a meromorphic map with irreducible projective image Z_j . Suppose that a dense open part of $q_j^{-1}(U_j)$, called the genuine frame locus, meets every fiber densely and that ϕ_j is holomorphic there. Let $G_j \subset Z_j$ be a countable union of algebraically constructible subsets.

Let $\Lambda_j \subset U_j$ be sets whose union contains a very general subset of Y , such that for $y \in \Lambda_j$ all genuine frames above y map into G_j . Then some Λ_j is not contained in a countable union of proper closed analytic subsets of Y , and for that index one of the constructible subsets making up G_j is dense in Z_j . In particular G_j contains a nonempty algebraic open subset of Z_j .

Proof. Some Λ_j is not contained in a countable union of proper closed analytic subsets; otherwise their union would be so contained. Resolve the graph of ϕ_j and shrink U_j , if necessary. This discards only a proper analytic subset of Y and preserves the stated properties of the general fibers. Write $G_j = \bigcup_i G_{ji}$. If no G_{ji} is dense, the inverse images

$$E_{ji} = \phi_j^{-1}(\overline{G_{ji}})$$

are proper closed analytic subsets of the resolved Q_j . For $y \in \Lambda_j$, the genuine frame fiber is covered by these subsets. Baire's theorem implies that one E_{ji} contains a relatively open part of the fiber. Irreducibility then implies that E_{ji} contains the whole compact fiber.

The locus

$$A_{ji} = \{y \in U_j : \dim(E_{ji} \cap q_j^{-1}(y)) \geq r_j\}$$

is analytic, by the fiber-dimension theorem for proper maps. It is proper: otherwise E_{ji} would contain the general fibers and hence all of Q_j . The same theorem before restriction to U_j shows that A_{ji} , together with $Y \setminus U_j$, is contained in a proper closed analytic subset of Y . This would place Λ_j in countably many proper closed analytic subsets, a contradiction. Finally, a dense constructible subset of an irreducible algebraic variety contains a nonempty open subset. $\square$

Lemma 5.2 (Analytic comparison with a stable family). *Let $h: V \rightarrow Y$ be a surjective holomorphic map with connected fibers, where V is a smooth compact Kähler manifold and Y is a smooth compact manifold in class $\mathcal{C}$. Let A be an effective rational SNC divisor on V , horizontal over Y , with coefficients strictly less than one. Suppose $K_F + A_F$ is big on very general fibers F of positive dimension. There are*

- (i) *a proper generically finite map $\tau: Y' \rightarrow Y$, with Y' smooth, compact, and in class $\mathcal{C}$;*
- (ii) *a surjective holomorphic map $\rho: Y' \rightarrow R$, with R smooth and projective;*
- (iii) *a projective stable family $\pi: (\mathcal{V}, \mathcal{A}) \rightarrow R$ of maximal variation, with klt general fiber;*

such that the main component of $V \times_Y Y'$ is bimeromorphic over Y' to $\mathcal{V} \times_R Y'$. Over a dense analytic open, this meromorphic comparison is the fiberwise log canonical model map for $K_F + A_F$, with the horizontal pushforward of A as boundary. The space R may be a point.

Proof. The construction has three outputs. Relative pluricanonical images produce algebraic parameter data even though the base is analytic. A classifying map then supplies a projective stable family after a finite base cover. Finally, an isomorphism cover matches that family with the relative image; fiberwise canonical models determine the horizontal discrepancy signs.

Relative images and frames. Set $\mathcal{L} = \mathcal{O}_V(K_{V/Y} + A)$, understood as a rational line bundle. For the factorial multiples m clearing its denominator, the sheaves

$$\mathcal{E}_m = h_* \mathcal{L}^{\otimes m}$$

are coherent. First restrict to a dense analytic open where h and the horizontal boundary strata are smooth, so the proper family and its line bundles are flat over the base. Generic constancy of the fiber section dimension and cohomology and base change then identify the fibers of $\mathcal{E}_m$ with the complete systems, separately for each m ; see [33, §7, nos. 4–6, Sätze 3–5] and its corrigendum [34]. After a modification of Y , the strict transform of $\mathcal{E}_m$, modulo torsion, is locally free. The evaluation map gives a relative meromorphic map into its projective bundle. Resolve this map and take its proper analytic image $\mathcal{Z}_m$. We also take the proper images of the horizontal boundary components and retain their divisorial cycle parts. Analytic flattening makes these embedded image and cycle data flat on a dense analytic open, with fixed Hilbert and cycle polynomials. These uses of coherent direct images and analytic flattening are independent of any minimal-model theorem; see [39].

Write $s_m = \text{rank } \mathcal{E}_m$ on this open; indices with $s_m = 0$ are omitted. Its projective frame bundle has a natural compactification

$$\overline{Q}_m = \mathbb{P}(\text{Hom}(\mathcal{O}_Y^{s_m}, \mathcal{E}_m)).$$

Here Y denotes the modification chosen for this degree; its map to the original base remains proper and bimeromorphic. The locus of invertible matrices modulo scalars is a principal $\text{PGL}(s_m)$ -bundle. A genuine frame embeds the corresponding fiber of $\mathcal{Z}_m$ in a fixed $\mathbb{P}^{s_m-1}$. There is therefore a holomorphic Hilbert–Chow parameter map on the genuine frame locus. The Hilbert spaces here are the algebraic Hilbert schemes of projective space, viewed analytically; equivalently one may use their Douady realization [24].

To extend this map meromorphically, use the existing compact relative image and the projective linear action in local trivializations. The graph of that action extends over the projective matrix space: inversion is given by the adjugate matrix. Apply this construction to the proper family $\mathcal{Z}_m$ and its marked cycles, take proper analytic images, and flatten by strict transform. One obtains a proper flat family of embedded subspaces over a modification of $\overline{Q}_m$. Its Hilbert–Chow map is holomorphic. The local graph constructions agree on the genuine frames and hence glue. Thus the original parameter map is meromorphic. After resolving it, its image is compact analytic in a fixed projective Hilbert–Chow space, and is consequently algebraic. Denote the irreducible image by Z_m . This construction uses a frame bundle, not a meromorphic trivialization of $\mathcal{E}_m$.

The good parameter locus. In Z_m , call an embedded pair (Z, D) good if it is klt and stable, has the prescribed boundary weights, has a complete nondegenerate embedding, and satisfies

$$(5.1) \quad \mathcal{O}_Z(1) \simeq \mathcal{O}_Z(m(K_Z + D)).$$

One may also impose the generic Hilbert and marking data. This is a countable union of algebraically constructible loci. Here is the parameter-space justification. Stratify the universal embedded families by their Hilbert polynomials, Cartier indices, and base-change loci. On each such stratum, Equation (5.1) can be imposed by parameterizing the map

$$\omega_Z^{[m]} \longrightarrow \mathcal{O}_Z(1)$$

with its prescribed divisor. These parameter spaces are of finite type. The stable-pair and klt conditions are constructible there, and their images are constructible by Chevalley’s theorem. This is the embedded construction used for stable log-variety moduli in [46, Section 6.A, especially Propositions 6.8 and 6.11 and Lemma 6.12]. In particular no assertion about analytic minimal models is hidden in the good locus.

Every very general fiber is projective: bigness makes it Moishezon, and it is Kähler. We use Moishezon’s projectivity criterion, which is also the smooth case of [49, Theorem 6]. The projective klt finite-generation theorem [11, Theorem 1.1 and its consequences] therefore gives its log canonical model. Every sufficiently large divisible pluricanonical degree embeds that model and satisfies Equation (5.1). Exclude at once the countably many generic base-change loci, relative-rank loci, and generic-degree loci for the maps constructed above. Let Λ_m consist of the remaining parameters for which the degree- m map actually constructs the fiber’s log canonical model. Their union contains a very general subset of Y , and all genuine frames over a point of Λ_m give good data. Apply Theorem 5.1. We obtain one fixed m for which Λ_m is not contained in a countable union of proper analytic subsets and good data hold on an algebraic open of Z_m .

For this degree the relative image map is generically birational. Indeed its generic finite degree agrees with its fiberwise degree outside the generic-degree exceptional set already excluded. Choose a point of Λ_m there; that degree is one. Retain Λ_m : its actual canonical models will determine the horizontal discrepancy signs at the end of the construction.

The invariant classifying map. Fix the dimension, the volume read from Equation (5.1), and a finite coefficient set containing the coefficients of A and closed under sums at most one. The good algebraic parameter locus has a classifying map to the projective coarse space M of stable pairs with these data. If an additional parameterization of the map in Equation (5.1) was used, its moduli graph descends: the embedded pair determines a unique isomorphism class, so the closure of that graph is generically a singleton over the good parameter locus. Restricting a dense algebraic open therefore gives the same rational classifying map on Z_m . It is invariant under change of projective frame.

Compose with the meromorphic frame parameter map. The resulting map $\overline{Q}_m \dashrightarrow M$ is constant on general frame fibers. Its graph has a proper analytic image in $Y \times M$. This image is generically a singleton over Y , so its projection to Y is bimeromorphic. It is the graph of a meromorphic map

$$\mu: Y \dashrightarrow M.$$

Let M_0 be the closure of its image; it is projective.

A stable parameter cover and the matching cover. Stable moduli supplies a finite parameter cover $S \rightarrow M$ from a normal scheme carrying a stable family [46, Corollary 6.19]. The coarse space M is projective [46, Corollary 7.3], so its finite cover S is projective as well. The moduli construction for the specified coefficient set is given in [46, Definition 6.2, Proposition 6.4, and Proposition 6.14]. Resolve μ , and normalize a component of the pullback of $S \rightarrow M$ dominating Y . Let R_0 be its image in S . The restricted stable family over R_0 has maximal variation, because $R_0 \rightarrow M_0$ is finite.

At this point the projective family exists over R_0 . Equality of coarse moduli points identifies its general fibers with those of the original analytic image, but does not identify the families. We construct a further generically finite cover carrying such an identification.

On a dense analytic open of the comparison base, embed the original flat image by the complete system of $\mathcal{O}_{Z_m}(k)$, and the pulled-back stable family by its km -adjoint system, for a common sufficiently large and divisible k . The two sheaves of sections are coherent direct images of compact families; generic cohomology and base change make them locally free after shrinking. On a common compact modification of the base, make their torsion-free strict transforms locally free, resolve the evaluation maps, and flatten the embedded image families and their weighted boundary cycles. This gives holomorphic relative Hilbert–Chow data. The modification concerns the base itself, not a space of choices of frames.

On every general good fiber, Equation (5.1) identifies the two polarizations. Thus the projective linear maps identifying the two embedded varieties and their weighted boundary cycles form a nonempty Isom set. This set is finite, since stable pairs have finite automorphism groups [46, Propositions 6.5 and 6.8]; completeness and nondegeneracy of the embeddings exclude additional ambient stabilizers. At this stage we use only the embedded analytic image and its polarization. Its $\mathbf{Q}$ -Gorenstein family structure will be transported from the stable family by the identification.

To compactify these Isom sets, work in local trivializations of the two section bundles. In the product of their projective parameter spaces with the projective matrix space, take the algebraic closure of the incidence relation for invertible matrices identifying the embedded varieties and weighted boundary cycles, after clearing the fixed coefficient denominators. Pull back this closure by the two Hilbert–Chow maps. Equivariance under changes of both frames glues the local incidence spaces into a proper analytic space in the projective bundle of homomorphisms between the section bundles; no meromorphic trivialization is used. Retain its compact irreducible components that dominate the base and meet the invertible locus, and normalize them. The invertible locus has the finite Isom fibers just described, so the resulting cover is proper and generically finite. Its tautological isomorphism identifies the generic families, including the stable family’s $\mathbf{Q}$ -Gorenstein structure; its graph therefore extends meromorphically over the compact base.

Finally resolve R_0 projectively and resolve the graph of the map from the comparison cover to it. Denote the resulting spaces by R and Y' . Stable families and their divisible relative adjoint line bundles commute with base change, with boundary understood by divisorial restriction [46, Remark 3.8]. Thus the pullback family over R remains stable; it has klt general fibers and maximal variation. All covers and modifications used above stay in class $\mathcal{C}$.

On a common resolution we can now compare the source adjoint divisor with the pullback of the stable relative adjoint divisor, which is rational Cartier. There are only finitely many horizontal prime coefficients in their difference. On fibers above Λ_m , away from the finitely many generic comparison exceptions, they are the log canonical model discrepancy coefficients. They are therefore nonnegative and supported on divisors exceptional over the stable model. The same statements hold horizontally in the family. Remove the images of the remaining vertical discrepancy divisors. On the resulting dense analytic open this is the log canonical model comparison, with its divisorial pushforward boundary. This proves the asserted meromorphic comparison without relative analytic finite generation. If M_0 is a point, the construction uses the fixed stable pair and R is a point. $\square$

5.2. Relative forms and vertical valuations. For a meromorphic section σ of $\mathcal{O}_V(bK_{V/Y})$, its horizontal pole bound bA means that $\text{ord}_E(\sigma) \geq -bA_E$ at every horizontal prime, using a local nonvanishing base volume to interpret relative forms. At a vertical prime over a divisor $D \subset Y$, write ω_D for a local nonvanishing logarithmic top form on the base with a simple pole along D , and no other pole at its generic point. We say that σ has the relative logarithmic bound at this prime if

$$(5.2) \quad \text{ord}_E(\sigma \wedge h^* \omega_D^b) \geq -b.$$

The notation $\wedge$ denotes the relative-times-base identification of the corresponding pluricanonical line bundles.

Lemma 5.3. *In the situation of Theorem 5.2, pull back a holomorphic section of a divisible multiple of $K_{V/R} + A$, and compare it with relative pluriforms on the main component of $V \times_Y Y'$. It satisfies the horizontal bound given by the pullback of A . Moreover, on a common resolution it satisfies Equation (5.2) at every prime lying over a divisor of any smooth modification of the comparison base.*

Proof. On general fibers, pullback from the log canonical model lies in the original log pluricanonical system. The effective canonical-model discrepancy on a common resolution therefore proves the horizontal bound.

For the vertical assertion, first fix a smooth modification of the comparison base and a prime divisor on it. The stable family and its divisible relative adjoint line bundles commute with this base change. Work transversely to a general point of that divisor. The resulting one-parameter family is still stable. This order of operations is what permits the same bound at divisors exposed only on a higher base model. Its total pair with the reduced central fiber added is log canonical. More explicitly, choose the disk with klt fibers away from its origin. Its total space is normal: flatness over the smooth disk and the S_2 property of the fibers give S_2 . A codimension-one point on the central fiber is a generic point of one of its reduced components. Its local ring is one-dimensional, and its quotient by the disk parameter is a field; its maximal ideal is consequently generated by that parameter, so this local ring is regular. Away from the central fiber the total space is normal, because its fibers are klt. Serre's criterion now gives normality. Compatibility of the relative adjoint divisor with the stable fiber identifies the adjunction different on the normalization of the central fiber with its stable boundary and conductor. The resulting pair is log canonical because the central fiber is stable; see [55, Lemma 2.10 and Corollary 2.11] for this normalization-and-conductor identity. The analytic Cartier-divisor inversion theorem [28,

Theorem 1.1] consequently applies. The argument also applies after any finite ramified disk change.

Let t be the disk parameter and let ξ denote the pulled-back relative section in degree b . Then $\xi \wedge (dt/t)^b$ is a section of the total log canonical multiple with the central fiber added. On a log resolution the log canonical discrepancy bound says that this form has pole order at most b at every vertical prime. A further common resolution with the original family preserves this bound. This is Equation (5.2). Smooth transverse parameters do not change any of these orders. $\square$

Lemma 5.4. *Under the hypotheses of Theorem 5.2, there is a positive integer b_0 and a finite-dimensional space*

$$\mathcal{S} \subset H_{\text{mer}}^0(V, \mathcal{O}_V(b_0 K_{V/Y}))$$

whose associated meromorphic map has rank $\dim F$ on general h -fibers, with the following properties:

- (i) *every member has horizontal pole bound $b_0 A$;*
- (ii) *every member satisfies the relative logarithmic bound at primes over base divisors;*
- (iii) *the same bound may be tested after any modification of the base, at a divisor exposed on that modification.*

The same assertions hold for the systems of products of members of $\mathcal{S}$.

Proof. Use Theorem 5.2. If $\dim R > 0$, the stable-family bigness theorem [46, Theorem 8.1 and Corollary 8.3] applies: the base is normal projective, the family is stable and has maximal variation, and its general fiber is klt. Thus $K_{\mathcal{V}/R} + \mathcal{A}$ is big. Choose finitely many sections of one sufficiently divisible multiple whose map is generically finite. Its image has dimension $\dim F + \dim R$, and its restriction to a general π -fiber has rank $\dim F$. If R is a point, choose such sections of the ample log canonical divisor of the fixed stable pair instead. Pullback and the meromorphic comparison preserve these image dimensions and give the pole bounds of Theorem 5.3.

Replace the comparison cover by a Galois cover and resolve the diagrams equivariantly. To do this analytically, first take its Stein factorization over Y ; its finite factor has the same cover over a dense open set. Over the finite étale locus, the Galois closure is a component of a sufficiently large fiber product of this finite factor. Take the corresponding compact component, normalize, and then resolve the graph of its map to the comparison cover. Equivariant resolution gives the required diagram; write G for its finite group. This construction uses finite analytic maps and does not require the meromorphic function field of Y to have transcendence degree $\dim Y$.

Let $\mathcal{B}$ be the graded $\mathbf{C}$ -algebra generated by the chosen finite set of relative sections and all its G -conjugates. Every homogeneous element has the required horizontal and vertical linear pole bounds: products add the orders and sums cannot decrease the lower bounds. The algebra is integral and finite over $\mathcal{B}^G$. Indeed each homogeneous generator satisfies its orbit polynomial with invariant coefficients, and invariants of a finite group acting on a finitely generated algebra are finitely generated. The corresponding map $\text{Proj } \mathcal{B} \rightarrow \text{Proj } \mathcal{B}^G$ is finite. Before taking invariants, the system contains the sections pulled back from $\mathcal{V}$, whose restriction to a general source fiber has image dimension $\dim F$. A finite map preserves the dimension of the image of that fiber, even though G may permute the fibers above one point of Y . Choose a Veronese degree for $\mathcal{B}^G$ generated by finitely many elements of that single degree. Their ratios therefore still have rank $\dim F$ on general fibers. Denote their pluricanonical degree by b_0 . The same finite quotient also preserves the total image dimension, which is at least $\dim F + \dim R$.

Use the natural identification of relative canonical bundles on the common smooth locus for every base change or modification. On the dense étale locus, invariant relative tensors therefore descend to V . They extend meromorphically across the remaining locus by finite analytic descent. The horizontal bounds descend because the comparison cover is generically étale along each horizontal divisor.

For completeness, the vertical order calculation is exact. Let E' lie over E , with ramification index e . For a degree- b_0 top pluriform α ,

$$(5.3) \quad \text{ord}_{E'}(\alpha_{\text{pullback}}) + b_0 = e(\text{ord}_E(\alpha) + b_0).$$

A logarithmic top form on the base pulls back to a nonvanishing logarithmic top form at a prime dominating its divisor. Apply Equation (5.3) to the relative form wedged with this base logarithmic form. The bound upstairs is therefore equivalent to Equation (5.2) downstairs. For (iii), pull the chosen Galois comparison cover and its original sections to a common refinement over the modified base. Lemma 5.3 applies to these same pullbacks at primes above the newly exposed base divisor; its bounds pass to their conjugates, products, and invariant combinations. The ramification calculation therefore gives the bound for the pullback of each fixed descended member of $\mathcal{S}$. Products add the order inequalities, and taking a full product system is a Veronese operation on its image, so products preserve the relative rank as well. $\square$

Lemma 5.5. *Let $h: V \rightarrow Y$ be a proper surjective holomorphic map of irreducible compact complex spaces, with Y smooth. If a prime divisor E of a smooth V does not dominate Y , there is a modification of the base and a dominating main-component model on which the strict transform of E maps onto a prime divisor of the modified base.*

Proof. Apply analytic flattening to h , by strict transform, and resolve the modified base. Pullback preserves the flatness of the intermediate family, whose fibers have the generic relative dimension d . This intermediate source is proper and bimeromorphic over V . On a resolution dominating it, the strict transform of E remains a divisor, because a proper modification of the smooth V is an isomorphism at the generic point of E . Its image in the flat intermediate source has dimension $\dim V - 1$: it still dominates E . The fiber-dimension bound consequently gives an image on the new base of dimension at least $\dim V - 1 - d = \dim Y - 1$. That image is proper, since its projection to Y is contained in $h(E)$, and properness makes it a closed analytic set. It is therefore a prime divisor. Further source resolutions retain the same generic image and do not change the valuation of E . $\square$

5.3. Proof of the addition proposition.

Proof of Theorem 3.2. We may suppose $\kappa(Y, K_Y + C) \geq 0$. Choose a Kähler modification $\nu: \tilde{V} \rightarrow V$, resolve the boundary, and put

$$\tilde{T} = \nu_*^{-1}T + \text{Exc}(\nu)_{\text{red}}.$$

By Theorems 2.1 and 2.3, pullback and descent identify the divisible log pluricanonical rings on the total spaces and on their very general fibers. In particular, fiberwise bigness is preserved, and $B(h \circ \nu, \tilde{T}) = B(h, T)$. Every prime over base codimension at least two still has coefficient one: it is either a strict transform of such a prime or is newly exceptional. Write (V, T) for this Kähler model; its sections descend to the original pair.

If $\dim F = 0$, the map is bimeromorphic. Pull back sections of $K_Y + C$. At a prime dominating a base divisor the multiplicity is one and the condition $C \leq B(h, T)$ gives the required pole bound. At an exceptional prime the log canonical discrepancy bound for (Y, C) , together with $T_E = 1$, gives the same conclusion. Thus pullback preserves all the base systems and proves the assertion. If Y is a point, the assertion is simply the assumed bigness. We may therefore assume both dimensions are positive.

Write T^h for the horizontal part of T . Choose a rational $\epsilon > 0$ sufficiently small and set

$$A = (1 - \epsilon)T^h.$$

Its coefficients are strictly less than one. To make $K_F + A_F$ big uniformly on very general fibers, consider $\epsilon = 1/j$ and all denominator-clearing degrees. Choose a fiber outside their countably

many generic base-change and relative-rank exceptional sets. Openness of the big cone on that projective fiber gives one choice of ϵ and one degree with full image dimension. The generic-rank choice makes the same degree have full rank on general fibers. Notice that no vertical coefficient has been lowered, and C has not been changed.

Apply Theorem 5.4 to A , and write σ for a member of its system in degree b , including any product degree. Let

$$\eta \in H^0(Y, \mathcal{O}_Y(b(K_Y + C))).$$

The relative-times-base product $\sigma \wedge h^*\eta$ is a meromorphic section of $\mathcal{O}_V(bK_V)$. We check that its poles are bounded by bT at every prime.

There is nothing further to check at horizontal primes, since $A \leq T^h$. Suppose next that E maps onto a prime divisor $D \subset Y$, and put

$$a_E = \text{ord}_E(h^*D).$$

Relative to a local nonvanishing logarithmic base volume ω_D , the section η vanishes to order at least $b(1 - C_D)$. The relative logarithmic bound gives

$$(5.4) \quad \text{ord}_E(\sigma \wedge h^*\eta) \geq -b + b a_E(1 - C_D).$$

The inf-multiplicity convention is precisely

$$B(h, T)_D = 1 - \max_{E \mapsto D} \frac{1 - T_E}{a_E}.$$

Consequently $C_D \leq B(h, T)_D$ implies

$$a_E(1 - C_D) \geq 1 - T_E,$$

and Equation (5.4) is at least $-bT_E$. This uses no integrality condition on the boundary multiplicities.

Finally, suppose $\text{codim}_Y h(E) \geq 2$. Expose a base divisor D' below the strict transform of E by Theorem 5.5, and resolve the diagrams. The pullback of η has at most a logarithmic pole along D' : this is the divisorial discrepancy inequality for the log canonical smooth pair (Y, C) . Property (iii) of Theorem 5.4 therefore gives order at least $-b$ at the strict transform of E . The order is unchanged, since source modifications are isomorphisms at the generic point of E . The assumption $T_E = 1$ supplies exactly this allowance. We have proved, at all prime divisors, that

$$(5.5) \quad \sigma \wedge h^*\eta \in H^0(V, \mathcal{O}_V(b(K_V + T))).$$

On a smooth space these divisorial bounds imply holomorphic extension of the corresponding log pluricanonical section.

Choose a degree in which the complete base system has image dimension $\kappa(Y, K_Y + C)$, and pass to a common multiple with b_0 in Theorem 5.4. The product system of relative sections still has image dimension $d = \dim F$ on a general fiber. The corresponding product of the chosen base system has image dimension $\kappa(Y, K_Y + C)$, since a Veronese embedding preserves its image. In this common degree write the nonzero relative generators as σ_i and the base generators as η_j . The total system in Equation (5.5) contains all $\sigma_i \wedge h^*\eta_j$. On the open set where σ_0 and η_0 are nonzero, ratios in this system include

$$\frac{\sigma_i}{\sigma_0}, \quad h^*\left(\frac{\eta_j}{\eta_0}\right).$$

Its image therefore maps onto the base Iitaka image, and over a general point of that image it contains the d -dimensional image of a general h -fiber. The dimension-of-fibers inequality gives image dimension at least $d + \kappa(Y, K_Y + C)$. This proves Equation (3.2) on the Kähler model. The initial log pluricanonical descent then proves it on the given smooth source. $\square$

6. RETURN TO THE ORIGINAL FIBRATION AND CONSEQUENCES

The previous sections supply the two positivity statements and the exact section comparison. We now apply them to the original fibration, keeping its base fixed while changing the intermediate Iitaka space.

Proof of Theorem 1.1. If either summand on the right is $-\infty$, the asserted inequality is automatic. Suppose henceforth that both are nonnegative, and put

$$k = \kappa(F, K_F + \Delta_F), \quad b = \kappa(f \mid \Delta).$$

Fixing the base. Use Section 2 and Theorem 2.4 to resolve the diagram, take Kähler models, and flatten before resolving the source. Use strict transforms plus reduced exceptional divisors at every source modification. Resolve the discriminant and boundary-stratum images on the base, then fix this smooth Kähler base Y . The source and very general fiber log section rings are unchanged, and every source prime mapped into codimension at least two in Y has coefficient one. Put

$$C = B(f, \Delta)$$

on this model. It has SNC support, and the definition of the invariant gives

$$(6.1) \quad \kappa(Y, K_Y + C) \geq b.$$

Theorem 2.5 preserves $C = B(f, \Delta)$ and the coefficient-one condition through all further changes over Y .

The relative Iitaka map and its Hodge line. Apply Theorem 2.6, obtaining

$$X \xrightarrow{g} W \xrightarrow{h} Y, \quad \dim W_y = k, \quad \kappa(G, K_G + \Delta_G) = 0.$$

Make W Kähler, resolve the diagram, and fix the current smooth source pair (X_0, Δ_0) as reference. Flatten over W and resolve the generator and boundary data as required by Theorem 2.7. Thus every prime mapped by g into codimension at least two is exceptional over X_0 , and on a dense smooth log open the relative systems have rank one in all sufficiently divisible degrees.

Apply Theorems 2.7 and 3.1. Their further modifications remain over Y ; Theorem 2.5 preserves the exceptional reference, while the variation and its parabolic line pull back from the common open. The relative Iitaka property persists by Theorem 2.6. On the resulting common model we obtain a rational line M , a rational SNC boundary T in $[0, 1]$, and a surjective map to a smooth projective variety:

$$(6.2) \quad p : W \longrightarrow S, \quad M = p^*P, \quad P \text{ nef}, \quad K_S + a_0P \text{ big for some } a_0 > 0,$$

such that

$$(6.3) \quad B(g, \Delta) \leq T \leq 1.$$

Compute T by the rank-one order formula on this final model. Parabolic pullback and Theorem 2.7 identify the divisible systems of $K_W + T + M$ with those on (X_0, Δ_0) , also relatively over Y , preserving all ratios.

Relative bigness and the base boundary. Set $D_W = K_W + T$. The systems defining the relative Iitaka image have image dimension k on very general f -fibers. Push these systems into $D_W + M$ using Theorem 2.7. Their ratios are unchanged, so the resulting systems have image dimension k on very general h -fibers, whose dimension is k . Consequently

$$(6.4) \quad (D_W + M)|_{W_y} \text{ is big for very general } y.$$

Here the systems are obtained by generic base change over an open patch of Y and extend on the smooth W_y by the relative section comparison. Adjunction identifies the restricted adjoint bundles.

By Theorem 2.12,

$$(6.5) \quad B(h, T) \geq C.$$

The same lemma gives coefficient one in T for every prime of W mapped into codimension at least two in Y . Thus both boundary hypotheses of Theorem 3.2 hold.

The hypotheses of Theorem 3.4 have now all been verified. Its conclusion is

$$(6.6) \quad \kappa(W, K_W + T + M) \geq k + \kappa(Y, K_Y + C).$$

The proof of that proposition uses a single nonzero section to cancel the ample perturbation and preserves the ratios defining this bound.

Finally, Theorem 2.7 lifts these systems by the relative generators, preserving their ratios and image dimensions. Untested g -contracted primes are exceptional over (X_0, Δ_0) , so the forms descend and extend there. By Theorem 2.1, they then descend to the original source. Combining (6.1) and (6.6) proves

$$\kappa(X, K_X + \Delta) \geq \kappa(F, K_F + \Delta_F) + \kappa(f | \Delta).$$

Bigness on a point means Iitaka dimension zero, and a zero-dimensional connected fiber has trivial relative line. If Y is a point the inequality is an identity; if the relative Iitaka fiber is a point its Hodge line is trivial. Together with the point-period-base case above and the initial $-\infty$ convention, these observations cover all dimensions and values of the two summands. $\square$

6.1. The logarithmic and ordinary inequalities. We first compare the invariant orbifold base with a prescribed logarithmic base. This makes the two classical specializations of Theorem 1.1 immediate, while retaining control of the base under modification.

Lemma 6.1 (Comparison with a logarithmic base). *Let $f : (X, \Delta) \rightarrow Y$ be a fibration of smooth compact complex manifolds, where Δ has rational coefficients in $[0, 1]$ and simple normal crossing support. Let D_Y be a reduced simple normal crossing divisor on Y . Suppose that every prime component of f^*D_Y has coefficient one in Δ . Then*

$$\kappa(f | \Delta) \geq \kappa(Y, K_Y + D_Y).$$

Proof. Choose a neat model as in Lemma 2.4, with $p : X' \rightarrow X$, $q : Y' \rightarrow Y$, $f' : X' \rightarrow Y'$, and $\Delta' = p_*^{-1}\Delta + \text{Exc}(p)_{\text{red}}$. We may arrange that $E_Y = (q^{-1}\text{Supp } D_Y)_{\text{red}}$ has simple normal crossings. Every prime of X' dominating a component of E_Y has coefficient one in Δ' : an exceptional prime has coefficient one by definition, and a nonexceptional prime is the strict transform of a component of f^*D_Y . Thus its weighted multiplicity is infinite, and the inf-multiplicity rule gives

$$B(f', \Delta') \geq E_Y.$$

Pullback by q takes logarithmic top forms with poles along D_Y to logarithmic top forms with poles along E_Y . In local coordinates this follows by pulling back each factor dz_i/z_i ; the same assertion holds for all tensor powers. Consequently, pullback embeds every plurilogarithmic system on (Y, D_Y) into the corresponding system of $K_{Y'} + B(f', \Delta')$, preserving all ratios of sections. The neat-model identity (2.2) now proves the claim. $\square$

Corollary 6.2 (Logarithmic subadditivity). *Let $f : X \rightarrow Y$ be a surjective morphism with connected fibers between smooth connected projective complex varieties. Let D_X, D_Y be reduced simple normal crossing divisors, with the zero divisor allowed, such that*

$$\text{Supp}(f^*D_Y) \subseteq \text{Supp}(D_X).$$

For a very general fiber F , put $D_F = D_X|_F$. Then

$$(6.7) \quad \kappa(X, K_X + D_X) \geq \kappa(F, K_F + D_F) + \kappa(Y, K_Y + D_Y).$$

The same statement holds for compact manifolds in Fujiki class $\mathcal{C}$ and holomorphic fibrations between them.

Proof. Theorem 1.1 applies to (X, D_X) , and Lemma 6.1 bounds its base term from below. If either term on the right of (6.7) is $-\infty$, the assertion follows from our convention for that value. $\square$

In the projective setting, this is the usual compactification form of the logarithmic Iitaka conjecture. Indeed, for a smooth quasi-projective variety U with a smooth projective compactification X and reduced simple normal crossing boundary $D_X = X \setminus U$, its logarithmic Kodaira dimension is $\bar{\kappa}(U) = \kappa(X, K_X + D_X)$ [41]. A dominant morphism $U \rightarrow V$ between smooth connected quasi-projective varieties, with geometrically connected general fiber, admits compatible such compactifications: compactify the graph and resolve the graph and boundaries. The Stein factor of the resulting morphism $X \rightarrow Y$ is finite birational over the normal variety Y , hence equals Y . Thus f has connected fibers, and $\text{Supp}(f^*D_Y) \subseteq \text{Supp}(D_X)$. For very general $v \in V$, $(X_v, D_X|_{X_v})$ compactifies U_v . Thus Corollary 6.2 gives $\bar{\kappa}(U) \geq \bar{\kappa}(U_v) + \bar{\kappa}(V)$.

Ordinary subadditivity also enters the Albanese reduction of [53]. That work combines it with separate nonvanishing and good-model arguments to prove log abundance in characteristic zero.

Corollary 6.3 (Ordinary subadditivity in characteristic zero). *Let k be an algebraically closed field of characteristic zero, and let $f : X \rightarrow Y$ be a surjective morphism with connected fibers between smooth connected projective k -varieties. If F is its geometric generic fiber, then*

$$(6.8) \quad \kappa(X) \geq \kappa(F) + \kappa(Y).$$

More generally, the logarithmic inequality of Corollary 6.2, with the same boundary hypotheses, holds over k , with the geometric generic fiber in place of a very general complex fiber.

Proof. Over $\mathbb{C}$, the ordinary assertion follows by taking $D_X = D_Y = 0$ in Corollary 6.2. In a projective family over $\mathbb{C}$, the Iitaka dimension of the geometric generic fiber equals that of a very general fiber. To see this, apply cohomology and base change in each positive divisible pluricanonical degree. After shrinking the base separately in each degree, the corresponding relative rational map also has constant fiberwise image dimension. Excluding the resulting countable union of proper closed subsets identifies all these image dimensions with those of the generic fiber.

For the field extension, let $K \subseteq K'$ be fields and let V be a smooth projective geometrically integral K -variety with a rational divisor D . For every m for which mD is integral, flat base change gives

$$H^0(V, \mathcal{O}_V(mD)) \otimes_K K' \simeq H^0(V_{K'}, \mathcal{O}_{V_{K'}}(mD_{K'})).$$

The complete linear system on the right is the scalar extension of the one on the left, so nonvanishing and image dimension are unchanged. Hence Iitaka dimension is invariant under extension of the ground field. Applying this observation to the generic fiber over the function field of the base proves the corresponding invariance for geometric generic fibers.

Descend f , its projective embeddings, and its boundary divisors to a finitely generated subfield $K \subset k$ over $\mathbb{Q}$. The descended varieties are smooth and geometrically integral; the boundary conditions descend as well. The equality $f_*\mathcal{O}_X = \mathcal{O}_Y$ descends by proper flat base change and faithful flatness, and therefore also holds after an embedding $K \hookrightarrow \mathbb{C}$. Apply the complex logarithmic inequality to this base change. The preceding invariance identifies its three Iitaka dimensions with those over k , proving both assertions. $\square$

6.2. Kodaira dimension along the core. A smooth compact orbifold pair (X, Δ) in class $\mathcal{C}$ is *special* if it has no meromorphic fibration onto a positive-dimensional base of orbifold general type; here general type means that the invariant base dimension equals the dimension of the base. Campana constructed its *core*

$$c_{(X, \Delta)} : (X, \Delta) \dashrightarrow C(X, \Delta),$$

whose very general orbifold fiber is special and whose orbifold base is of general type, unless it is a point [16, Theorem 9.1]. In particular, the core has a point base exactly when (X, Δ) is special. Fiber properties of the core and the tower below refer to Campana's stable orbifold fibers: they are computed on the suitable equivalent holomorphic representatives furnished by these constructions [16, Definition 2.49 and §7.1]. A neat representative $g : (Z, \Gamma) \rightarrow T$ in the sense of Section 2 is *strictly neat* if its inf-multiplicity boundary on the smooth base is SNC. It is *high* if g is an orbifold morphism to this base: for every base prime D and source prime E with $a_E = \text{ord}_E(g^*D) > 0$, including primes contracted by g ,

$$a_E m_\Gamma(E) \geq m_{B(g, \Gamma)}(D).$$

These are the model conditions of [16, Definition 3.8 and Proposition 3.10]. The high condition is not automatic: the inf-multiplicity base tests divisors dominating base primes, whereas an orbifold morphism must also control primes contracted into higher codimension [16, Remark 3.3(2)]. For the core and its tower we choose compatible strictly neat and high representatives supplied by Campana's constructions [16, Proposition 3.15 and Corollary 4.14]. The section comparisons of Section 2 do not assert that arbitrary model changes preserve every stable-fiber property.

Corollary 6.4 (Kodaira dimension along the core). *Let (X, Δ) be a smooth compact orbifold pair in Fujiki class $\mathcal{C}$, with rational simple normal crossing boundary coefficients in $[0, 1]$. Let $C = C(X, \Delta)$ be its core base and (G, Δ_G) a very general orbifold fiber. Then*

$$(6.9) \quad \kappa(X, K_X + \Delta) = \kappa(G, K_G + \Delta_G) + \dim C.$$

If $\kappa(X, K_X + \Delta) \geq 0$, then

$$0 \leq \dim C \leq \kappa(X, K_X + \Delta),$$

and equality in the second inequality holds precisely when $\kappa(G, K_G + \Delta_G) = 0$. In particular, $\kappa(X, K_X + \Delta) = 0$ implies that (X, Δ) is special.

Proof. Resolve the core map, using the boundary convention of Lemma 2.1. The total and fiber Iitaka dimensions are preserved. Theorem 1.1 gives the lower bound in (6.9), since the invariant base dimension is $\dim C$.

For completeness, the reverse bound is the elementary upper addition inequality. For each divisible m , let ϕ_m be the meromorphic map defined by $|m(K_X + \Delta)|$, when this system is nonempty. The map (c, ϕ_m) has image dimension at most $\dim C + \kappa(G, K_G + \Delta_G)$: on a very general c -fiber its second component is given by a subspace of $H^0(G, m(K_G + \Delta_G))$. Since projection from this image onto the image of ϕ_m is surjective, the same upper bound holds for $\dim \phi_m(X)$. Taking the maximum over m proves the desired inequality. If the fiber Iitaka dimension is $-\infty$, restriction to a very general fiber shows that no global pluricanonical section can be nonzero, so both sides of (6.9) are $-\infty$.

Finally, if the total Iitaka dimension is nonnegative, a nonzero section restricts nontrivially to a very general core fiber. Its Iitaka dimension is therefore nonnegative. The dimension bound and its equality case follow from (6.9). When the total dimension is zero, the core base is a point, which is equivalent to specialness. $\square$

These statements recover, for the core, Campana's general-type-base addition theorem and its consequences [16, Theorems 6.3, 6.7, and 6.8]. The existence of the core and the implication from Kodaira dimension zero to specialness are established results of his theory.

6.3. The core tower. For a smooth orbifold pair (Z, Θ) , the notation $\kappa_+(Z, \Theta) = -\infty$ means that every meromorphic fibration onto a positive-dimensional base has invariant orbifold Kodaira dimension $-\infty$; the condition is vacuous for a point. Equivalently, one may test all dominant meromorphic maps: in Stein factorization, the finite orbifold canonical-divisor inequality shows that passing to the connected-fiber factor cannot decrease the invariant base dimension. Indeed, over a divisor with multiplicity m_D , a prime with ramification index e and orbifold multiplicity m_E satisfies $m_D \leq em_E$; the canonical-divisor difference has coefficient $e/m_D - 1/m_E \geq 0$. The κ -rational quotient is a meromorphic fibration $r : (Z, \Theta) \dashrightarrow R$ with $\kappa(r|_{\Theta}) \geq 0$ whose very general stable orbifold fiber has $\kappa_+ = -\infty$ [16, Definition 5.23 and Corollary 6.14]. Its *stable orbifold base* is the inf-multiplicity base on the compatible representatives just described [16, Proposition 3.15, Corollary 4.14, and §7.1].

Corollary 6.5 (Campana's decomposition of the core). *Let (X, Δ) be as in Corollary 6.4, and set $n = \dim X$. Starting from (X, Δ) , successively take the κ -rational quotient r and the Moishezon–Iitaka fibration M of its stable orbifold base. At each iteration choose such compatible representatives and equip the new base with the stable inf-multiplicity orbifold structure of the composite. The composite after at most n iterations is the core, up to bimeromorphic equivalence of fibrations:*

$$c_{(X, \Delta)} = (M \circ r)^n.$$

The positive-dimensional very general stable orbifold fibers of the r -steps have $\kappa_+ = -\infty$, and those of the M -steps have Kodaira dimension zero. In particular, for a special pair this tower ends at a point.

Proof. Theorem 1.1 establishes the hypothesis of Campana's Corollary 6.14, which supplies the κ -rational quotient. Its scope is exactly that used here: smooth compact class- $\mathcal{C}$ pairs with rational coefficients in $[0, 1]$ [16, Conjecture 6.1 and Remark 6.2]. Its stable base has nonnegative Kodaira dimension, so M is defined and has very general orbifold fiber of Kodaira dimension zero. The new smooth base remains in class $\mathcal{C}$, with rational boundary coefficients in $[0, 1]$. Thus the construction can be repeated.

Campana's Lemma 10.1 states that each composite $M \circ r$ has special general orbifold fiber and decreases dimension unless its source pair is of general type. His Theorems 10.2–10.3 identify the resulting stabilized composite with the core. Their sole conjectural input is the existence of the κ -rational quotient, now supplied above. Strict dimension decrease allows at most n nontrivial iterations; identity steps give the displayed formula. A point is fixed throughout. These are precisely the decomposition and fiber assertions of [16, Lemma 10.1, Theorems 10.2–10.3, and Corollary 10.4]. $\square$

7. MINIMAL ROOT COVERS FOR REDUCED BOUNDARIES

For a smooth projective pair of logarithmic Kodaira dimension zero with reduced SNC boundary, the minimal cyclic cover has a one-dimensional space of logarithmic pluriforms in every positive degree. This strengthens the character-line statement of Proposition 2.7. We prove the refinement together with its exact Hodge and divisorial normalization. It gives a pure integral realization of the entire highest line and fixes the normalization used in further relative-Iitaka constructions. The normalization lemmas below compare the minimum over actual fiber components with the threshold over all divisorial valuations. They show that a zero coefficient of the normalized base boundary yields a multiplicity-one component, and that later permitted modifications preserve the complete section spaces. These are the additional normalization statements used by the variation companion.

Lemma 7.1 (The cyclic cover of a logarithmic Kodaira-zero pair). *Let (J, D) be a smooth geometrically integral projective pair over a field K of characteristic zero. Assume that D is*

reduced and has simple normal crossings and that $\kappa(J_{\overline{K}}, K_{J_{\overline{K}}} + D_{\overline{K}}) = 0$. Choose the least positive integer p for which $H^0(J, p(K_J + D)) \neq 0$, a nonzero section ω in that space, and put

$$E = \operatorname{div}(\omega) + pD, \quad \Delta = D - \frac{1}{p}E.$$

Write $\omega = b\eta^{\otimes p}$ in a rational canonical frame, and let $\tau : N \rightarrow J$ be the normalization in $K(J)[t]/(t^p - b)$. Let $\rho : \tilde{N} \rightarrow N$ be a log resolution, and set

$$P = \operatorname{Supp}(\Delta^{\perp 1}), \quad D_{\tilde{N}} = ((\tau \circ \rho)^{-1}P)_{\operatorname{red}}.$$

Then N is geometrically integral, the rational top form $\Omega = t\tau^*\eta$ extends logarithmically to $(\tilde{N}, D_{\tilde{N}})$, and

$$H^0(\tilde{N}, m(K_{\tilde{N}} + D_{\tilde{N}})) = K \Omega^m \quad (m \geq 1).$$

Proof. Every nonzero plurilogarithmic section space on J is one-dimensional: two independent sections of one degree would define a nonconstant rational map. Consequently $\kappa(E) = 0$, and every effective integral divisor supported on E also has exactly one section, since it is bounded above by a multiple of E . These assertions and the minimal index p are unchanged by field extension, by flat base change for global sections.

Over $\overline{K}(J)$, reducibility of $T^p - b$ would make b an ℓ th power for some prime $\ell \mid p$. The corresponding root of ω would be a nonzero logarithmic pluriform of degree p/ℓ , since its logarithmic zero divisor is E/ℓ . This contradicts minimality. Thus N is geometrically integral.

At a prime Q of N above a prime R of J , write

$$a = \operatorname{coeff}_R E, \quad d = \operatorname{coeff}_R D \in \{0, 1\}, \quad e = \operatorname{ord}_Q(\tau^*R).$$

The ramification formula for top differentials gives

$$(7.1) \quad \operatorname{ord}_Q \Omega = \frac{ea}{p} - ed + e - 1.$$

If $a = 0$, the cover is unramified at R , and this order is $-d$. If $a > 0$, then $q = ea/p$ is a positive integer: the integrality follows from $t^p = b$ and $d \in \mathbf{Z}$. The order is $q - 1 \geq 0$ when $d = 1$, and $q + e - 1 \geq 0$ when $d = 0$. Thus the divisorial poles are exactly the reduced inverse image $D_N = (\tau^{-1}P)_{\operatorname{red}}$.

We also check extension across the exceptional divisors of ρ . The subpair (J, Δ) is sub-log-canonical and is sub-plt outside P . Indeed, its positive part is an SNC boundary bounded by D , with all coefficients strictly below one outside P ; subtracting an effective divisor cannot worsen discrepancies. Define the crepant finite pullback by $K_N + \Delta_N = \tau^*(K_J + \Delta)$. With the compatible canonical trivialization, $\Delta_N = -\operatorname{div}(\Omega)$. For a prime divisor V over N , let its restricted valuation on $K(J)$ be $e_V v$. The same ramification formula gives

$$1 + \operatorname{ord}_V \Omega = e_V A_{J, \Delta}(v),$$

where A denotes log discrepancy. The right-hand side is nonnegative, and is positive if the center lies outside P . Since the left-hand side is integral, Ω has at most a simple pole above P and is regular elsewhere. This proves the asserted logarithmic extension.

Finally let $H_N = \operatorname{div}_N(\Omega) + D_N$. Equation (7.1) gives the sharper bound

$$0 \leq H_N \leq \tau^*E.$$

For completeness, when $a > 0$ and $d = 1$ its coefficient is $q - 1 \leq ea$; when $d = 0$, it is $q + e - 1 \leq q + p - 1 \leq pq = ea$, because $e \leq p$ and $q \geq 1$. At $a = 0$ its coefficient is zero. Writing a logarithmic m -canonical form as $u\Omega^m$ and testing its orders at the primes of the finite normalization gives

$$\operatorname{div}_N(u) + mH_N \geq 0, \quad u \in H^0(N, \mathcal{O}_N(m\tau^*E)).$$

Finite pullback preserves Iitaka dimension, so $\kappa(N, \tau^*E) = \kappa(J, E) = 0$. Since τ^*E is effective and N is geometrically integral, the last space is K . Conversely Ω^m is a logarithmic pluriform for every $m \geq 1$, proving the equality. $\square$

Corollary 7.2 (An integral pure realization of the moduli line). *Let $x : X \rightarrow W$ be a projective fibration between smooth complex projective varieties, with geometrically integral generic fiber J . Let D_X be a reduced SNC boundary whose restriction D_J satisfies the hypotheses of Lemma 7.1. Let M_W be the rational Hodge divisor of the rank-one reduction, with its rational frame and parabolic orders as in Equation (2.14). On sufficiently high smooth models, there is a polarizable integral pure variation of Hodge structures on a dense smooth open $W^\circ \subset W$ whose top nonzero Hodge filtration piece is a line, and*

$$M_W \sim_{\mathbf{Q}} \text{the rational canonical extension of that line.}$$

In the canonical-bundle-formula normalization, this is the moduli divisor of the auxiliary generic subpair $\Delta_J = D_J - E_J/p$, which may have negative coefficients.

Proof. Let s be the minimal-degree logarithmic generator on the generic fiber, and choose a rational relative canonical frame η on X . The rational function $s/\eta^{\otimes p}$ defines one cyclic cover over a dense open of W , not merely a collection of local character covers. Spread its resolution and pole boundary over a smaller dense open W° so that they form a smooth projective family of SNC pairs. Write $r = \dim J$, and let $u : \tilde{X}^\circ \setminus D_{\tilde{X}}^\circ \rightarrow W^\circ$ be the complementary open family. Its geometric variation

$$\mathbb{V}_{\mathbf{Z}} = R^r u_* \mathbf{Z} / \text{torsion}$$

is admissible and graded-polarizable. Logarithmic Hodge theory identifies

$$F^{r+1} \mathbb{V} = 0, \quad F^r \mathbb{V}_{\mathcal{O}} = \tilde{x}_* \omega_{\tilde{X}^\circ/W^\circ}(D_{\tilde{X}}^\circ).$$

Lemma 7.1 makes this entire top piece a line. Strictness for the weight filtration therefore singles out one weight w with

$$F^r W_{w-1} \mathbb{V} = 0, \quad F^r W_w \mathbb{V} = F^r \mathbb{V}, \quad F^r \text{Gr}_w^W \mathbb{V} \simeq F^r \mathbb{V}.$$

The rational variation $\text{Gr}_w^W \mathbb{V}$ is polarizable. Intersecting its rational weight filtration with $\mathbb{V}_{\mathbf{Z}}$ and removing torsion supplies an integral lattice; after rescaling, its rational polarization is integral. Thus this pure variation has precisely the required top line.

After a finite cover making local monodromies unipotent, the canonical extensions have locally free double gradeds for the Hodge and weight filtrations. The displayed identification of top lines consequently extends across the boundary; see [29, Proposition 3.12]. Descent gives the same identification for their rational canonical extensions.

The global rational frame of this entire top line is $s^{1/p}$. Equation (2.14) computes its parabolic order at every prime divisor P as β_P , the order of the same rational frame in M_W . Equality of these divisorial orders identifies the rational extensions. In particular, it identifies M_W itself, with no additional factor of p . We check the rank condition for the canonical-bundle-formula comparison. On a log resolution $\mu : J' \rightarrow J$, let $R' = \lceil A^*(J, \Delta_J)_{J'} \rceil$, where A^* omits the log-discrepancy-zero components of the discrepancy divisor. The subpair is sub-log-canonical, so $R' \geq 0$. At the strict transform of each original prime its coefficient is zero if $0 \leq \Delta_J \leq 1$, and is $\lceil -\Delta_J \rceil$ otherwise. Thus $0 \leq \mu_* R' \leq E_J$. Testing original primes gives $H^0(J', \mathcal{O}_{J'}(R')) \subseteq H^0(J, \mathcal{O}_J(E_J)) = \mathbf{C}(W)$, and constants give equality. This is the required rank-one condition. Extending the generic subpair by $-p^{-1} \text{div}_{K_{X/W}}(s)$ makes its log canonical divisor rationally equivalent to $x^* K_W$. The generic sub-lc, relative triviality, and rank conditions of [7, Definition 6.18] therefore hold. Adding a base pullback to normalize vertical orders changes the total base term and the discriminant equally, leaving the moduli divisor unchanged.

The least positive index trivializing $K_J + \Delta_J$ is p . Indeed, a smaller such index q would produce a rational q -canonical form with logarithmic zero divisor $qE_J/p \geq 0$, contradicting the minimality of p . Thus the cyclic field extension in Construction-Definition 6.23 of [7] is the one used above.

For its boundary comparison, put $P_J = \text{Supp}(\Delta_J^{-1})$ and first make a common crepant log resolution of the auxiliary subpair and its root cover. The coefficient-one support on the crepant model is contained in the total inverse image of P_J : outside P_J the auxiliary subpair is sub-plt. Every additional component allowed by our pole boundary has positive log discrepancy, so the ramification discrepancy formula in Lemma 7.1 makes Ω regular there. The top-form space for the BFMT pole boundary therefore injects into the top-form space for our boundary and contains Ω ; both are the line generated by Ω . After spreading over a common smooth base open, inclusion of the open complements induces a morphism of their admissible geometric mixed variations. This morphism is strict for the Hodge and weight filtrations, including their limiting filtrations after a finite cover making boundary monodromy unipotent. It therefore identifies the top lines in the same pure weight grade. Functoriality of canonical extension gives the same identification across the boundary and hence for the parabolic rational lines. Theorem 6.28 of [7] now identifies this Hodge divisor with the moduli divisor using one common positive multiple on both sides. This comparison permits negative coefficients in the generic subpair; it does not use the separate semiample theorem requiring generic effectiveness. $\square$

Remark 7.3. The hypothesis that D is reduced is essential to Lemma 7.1. For five distinct points on $\mathbf{P}^1$, the rational boundary $D = \frac{2}{5}(P_1 + \cdots + P_5)$ satisfies $K_{\mathbf{P}^1} + D \sim_{\mathbf{Q}} 0$. Its index-five root cover has genus six and empty pole boundary. Thus its entire top-form space has dimension six. The refinement therefore applies to reduced boundaries; the character-line construction of Section 2.6 is used for general rational orbifold boundaries.

7.1. The original relative orders.

Lemma 7.4 (Normalization of the rank-one boundary). *Let $x : X' \rightarrow W$ be a projective fibration on smooth models, let D' be an effective reduced SNC divisor, and suppose that its geometric generic fiber (J, D_J) has logarithmic Kodaira dimension zero. Fix a nonzero relative rational p -canonical form s generating $H^0(J, p(K_J + D_J))$. At a prime P of W set*

$$a_E = \text{ord}_E(x^*P), \quad r_E = p^{-1} \text{ord}_E^{pK_{X'/W}}(s), \quad d_E = \text{coeff}_E(D'), \quad t_P = \min_{E \mapsto P} \frac{r_E + d_E}{a_E}.$$

Here the minimum is over actual components on the chosen source model. Choose that model so that the generator divisor, boundary and inverse image of P have SNC support over the generic point of P ; later source modifications carry strict boundary plus reduced exceptional boundary. The coefficients in the rank-one comparison satisfy

$$\alpha_P = t_P, \quad \beta_P = \inf_{E \mapsto P} \frac{1 + r_E}{a_E} - 1, \quad T_P = 1 - c_P, \quad c_P := \inf_{E \mapsto P} \frac{1 + r_E}{a_E} - t_P.$$

*The infima allow divisorial valuations on higher smooth source models. Equivalently, c_P is the log canonical threshold of x^*P for the normalized auxiliary subpair*

$$\Delta^{\text{aux}} = -p^{-1} \text{div}_{K_{X'/W}}(s) + x^* \sum_Q t_Q Q.$$

If $T_P = 0$, an actual component attaining t_P has $a_E = 1$, $d_E = 0$, and attains the threshold infimum as well.

Proof. Choose a local frame η of K_W that does not vanish at the generic point of P . The canonical line-bundle identification

$$\mathcal{O}_{X'}(pK_{X'/W}) \otimes x^* \mathcal{O}_W(pK_W) \simeq \mathcal{O}_{X'}(pK_{X'})$$

identifies the order of s with the order of $s \wedge x^* \eta^{\otimes P}$. Thus the quantity l_E in the rank-one order formula is exactly r_E . This uses the actual relative canonical line bundle: the Jacobian contribution is already present in r_E , so no ramification term is added or subtracted here. Consequently its two order formulas give

$$\alpha_P = t_P, \quad \beta_P = \min_E \frac{r_E + 1 - a_E}{a_E}.$$

Put

$$\Delta_0 = -p^{-1} \operatorname{div}_{K_{X'/W}}(s), \quad \lambda_P = \min_{E \rightarrow P} \frac{1 + r_E}{a_E},$$

where the minimum is over actual vertical components. The indicated SNC preparation and the generic logarithmic bound give Δ_0 SNC support and horizontal coefficients at most one. In $\Delta_0 + \lambda_P x^* P$, each vertical coefficient is $-r_E + \lambda_P a_E \leq 1$. Thus this subpair is sub-log-canonical over the generic point of P , and every higher divisorial valuation satisfies

$$1 + r_E - \lambda_P a_E \geq 0.$$

An actual component attaining λ_P gives equality. Hence the infimum over all such valuations equals λ_P , and $\beta_P = \lambda_P - 1$.

For completeness, the first minimum also remains unchanged on higher source models. A coefficient realizing t_P makes $t_P a_E - r_E$ no greater than the original boundary coefficient at every actual component over P . Equivalently, after clearing denominators, the normalized generating form satisfies every logarithmic order test over the generic point of P . Its pullback remains logarithmic with strict plus reduced exceptional boundary. No new valuation can lower the first minimum, while an old minimizing valuation persists.

At any divisorial valuation above the generic point of P , the crepant coefficient of Δ^{aux} is $-r_E + a_E t_P$. The threshold formula therefore gives

$$\operatorname{lct}_{\eta_P}(X', \Delta^{\text{aux}}; x^* P) = \inf_E \frac{1 + r_E - a_E t_P}{a_E} = c_P.$$

Since $T_P = \alpha_P - \beta_P$, we obtain $T_P = 1 - c_P$. Finally, if $T_P = 0$ and E attains the first minimum, then

$$1 = c_P \leq \frac{1 + r_E}{a_E} - t_P = \frac{1 - d_E}{a_E} \leq 1.$$

All terms are equal. Since a_E is a positive integer and $d_E \in \{0, 1\}$, this forces $a_E = 1$ and $d_E = 0$, as claimed. $\square$

Lemma 7.5 (Permitted models after normalization). *Suppose that (W, T) is smooth with effective SNC rational boundary and that the moduli divisor M is a rational Cartier divisor on W . For a smooth modification $\mu : W' \rightarrow W$, put*

$$T' = \mu_*^{-1} T + \operatorname{Exc}(\mu)_{\text{red}}, \quad M' = \mu^* M,$$

and assume the resulting boundary is SNC. Then pullback identifies the section spaces of every sufficiently divisible multiple of $K_W + T + M$ and $K_{W'} + T' + M'$. At unchanged codimension-one points the normalization in Lemma 7.4 is preserved. Every new exceptional prime has coefficient one in T' .

Proof. Log canonicity of (W, T) gives

$$K_{W'} + T' + M' = \mu^*(K_W + T + M) + E,$$

where E is effective and μ -exceptional. For divisible m , normality gives

$$\mu_* \mathcal{O}_{W'}(mE) = \mathcal{O}_W.$$

The projection formula therefore identifies the section spaces. The modification is an isomorphism at each old generic divisorial point, so the old local order calculations are unchanged. The final assertion is the definition of T' . $\square$

Appendix A gives complementary projective proofs of the section comparisons and further calculations with the auxiliary subpair on higher models.

APPENDIX A. SUPPLEMENTARY PROJECTIVE CALCULATIONS

The reduced-boundary construction of Section 7 admits complementary projective proofs and calculations. We first give divisorial proofs of the section comparisons and recover the fixed-source descent by extracting valuations on the base. The fixed logarithmic zero divisor then gives another discrepancy-sheaf test, and a toroidal calculation proves logarithmic extension of the root form. The final subsection tracks the auxiliary subpair when the minimum-rule divisor is recomputed on a higher base model. This last operation is distinct from the prescribed boundary modification of Lemma 7.5: its discriminant is recomputed, rather than defined as the strict boundary plus the reduced exceptional divisor. Throughout these calculations, modifications of the original source carry strict boundary plus reduced exceptional boundary.

A.1. Divisorial sections and relative Iitaka systems. The following elementary comparisons fix the complete systems and the original models used in the normalization. All varieties in this subsection are projective over $\mathbf{C}$, all rational divisor degrees are sufficiently divisible, and the very general fibers are taken integral. The logarithmic modification identity, also with a pulled-back rational twist, is Theorem 2.1.

Lemma A.1 (Divisorial section test). *Let V be normal and D an integral Weil divisor. A rational section belongs to $H^0(V, D)$ if and only if it satisfies the corresponding order inequality at every prime divisor of V . If $\pi : V' \rightarrow V$ is finite surjective with V' normal and D is Cartier, this test on V' needs only primes lying over prime divisors of V . If π is Galois, then*

$$H^0(V', \pi^* D)^{\text{Gal}(V'/V)} = H^0(V, D).$$

Proof. The first assertion is the description of a rank-one reflexive sheaf as the intersection of its lattices at codimension-one points. For a finite morphism between normal varieties, a prime divisor upstairs lies over a prime divisor downstairs. The last assertion follows from $(\pi_* \mathcal{O}_{V'})^{\text{Gal}(V'/V)} = \mathcal{O}_V$ and the projection formula. $\square$

Lemma A.2 (Finite pullback). *If $\pi : V' \rightarrow V$ is a dominant generically finite morphism of normal projective varieties and D is a rational Cartier divisor on V , then*

$$\kappa(V', \pi^* D) = \kappa(V, D).$$

If $\kappa(V, D) = 0$, every nonzero space $H^0(V, mD)$ has dimension one.

Proof. Stein factorization reduces the first assertion to a finite morphism, since a proper birational morphism onto a normal variety has direct image $\mathcal{O}_V$. After replacing D by a multiple, use the natural inclusion of graded section rings

$$R(V, D) \subseteq R(V', \pi^* D).$$

Each homogeneous section upstairs is integral over the ring downstairs. Indeed its monic characteristic polynomial in the finite function-field extension has coefficients which are rational sections of the appropriate multiples of D ; they are regular in the required lattices at every prime divisor, and hence are global sections by Lemma A.1. Thus the two nonzero graded rings have the same transcendence degree, which gives equality of Iitaka dimensions. If the ring downstairs has no positive-degree section, the same norm argument excludes one upstairs. Finally, two independent

sections in one degree have a nonconstant ratio and define a positive-dimensional rational image. This proves the last assertion. $\square$

Lemma A.3 (Logarithmic pullback). *Let $\rho : V \rightarrow Y$ be a dominant generically finite morphism of smooth projective varieties. Let D_Y be a reduced snc divisor, and let B be a boundary on V satisfying $B \geq (\rho^* D_Y)_{\text{red}}$. Assume that the boundary supports on the chosen models are snc. Then pullback of rational pluricanonical forms gives*

$$H^0(Y, m(K_Y + D_Y)) \longrightarrow H^0(V, m(K_V + B))$$

injectively, for divisible m .

Proof. A logarithmic differential on a smooth snc pair pulls back to a logarithmic differential along the reduced inverse image. In local coordinates this follows from $d(\rho^* z)/\rho^* z$, whose divisorial poles are simple. Taking top exterior powers and then tensor powers proves the order inequalities. Equivalently, at a prime of ramification index e over a boundary component, the Jacobian contributes $e - 1$, and the coefficient-one boundary supplies the remaining one. At exceptional-image primes the same logarithmic differential calculation applies. Injectivity follows from dominance and separability. $\square$

Lemma A.4 (Relative Iitaka fibration). *Let $p : V \rightarrow B$ be a dominant projective morphism, with V smooth projective, and let D be a rational divisor whose restriction to the geometric generic fiber has Iitaka dimension $k \geq 0$. After birational modifications there is a relative Iitaka fibration*

$$V' \xrightarrow{q} U \longrightarrow B$$

in which q has geometrically connected generic fiber, such that

$$\dim U - \dim B = k, \quad \kappa(G, D'|_G) = 0$$

for a very general fiber G of q . Here $D' = \mu^ D$, where $\mu : V' \rightarrow V$ is the modification. When $D = K_V + B_V$, one may instead take the full log divisor from Lemma 2.1.*

If $D' = K_{V'} + B_{V'}$ and the models are log smooth, then $D'|_G \sim_{\mathbf{Q}} K_G + B_G$.

Proof. Apply the Iitaka fibration theorem over the function field of B [40, Section 6, Remark 1], take the relative algebraic closure of its section field in $\mathbf{C}(V)$, and spread out. Resolve the resulting rational maps and take their Stein factorizations. The dimension of the relative section field is k ; the restriction of the full divisor to the Iitaka fiber has Iitaka dimension zero by the Iitaka fibration theorem. For the logarithmic replacement, the proof of Lemma 2.1 gives the sheaf identity

$$\mu_* \mathcal{O}_{V'}(m(K_{V'} + B_{V'})) = \mathcal{O}_V(m(K_V + B_V))$$

in every sufficiently divisible degree. Pushing to the original base identifies the relative section systems, even if there are no global sections. Apply the same full-divisor Iitaka theorem to the replacement divisor; its relative section field identifies its fibration with the one already constructed and gives the asserted Iitaka dimension zero on the fiber. Generic smoothness and adjunction give the last assertion. The dimensions at very general complex fibers agree with the geometric generic values: for each divisible degree use cohomology and base change, and remove the union of the exceptional closed sets over the countably many degrees. $\square$

Lemma A.5 (Easy addition). *Let $p : V \rightarrow B$ be dominant and projective, with V smooth projective and integral very general fiber G , and let D be a rational divisor on V . Then*

$$\kappa(V, D) \leq \dim B + \kappa(G, D|_G).$$

If D has a nonzero global section, its restriction to a general fiber is nonzero.

Proof. A nonzero section cannot vanish identically on all fibers over a dense open subset. For each divisible m with sections, let ϕ_m be its rational map. The image of (p, ϕ_m) has dimension at most $\dim B + \kappa(G, D|_G)$, and dominates the image of ϕ_m . Taking the maximum over m proves the inequality. If the fiber term is $-\infty$, the last assertion implies that the total term is $-\infty$ as well. $\square$

A.2. Restricted valuations and the fixed original source. Let $p : H \rightarrow I$ be a dominant morphism of smooth projective varieties, and fix the model H . A divisorial valuation v of $\mathbf{C}(H)$ whose restriction to $\mathbf{C}(I)$ is nontrivial restricts to a positive integral multiple of a divisorial valuation of $\mathbf{C}(I)$. To see this, put $n = \dim H$, $m = \dim I$, and denote the restricted valuation by w . The value groups have rational rank one with finite-index inclusion. The relative residue-transcendence-degree inequality gives

$$\mathrm{trdeg}_{\kappa(w)} \kappa(v) \leq n - m.$$

As $\mathrm{trdeg}_{\mathbf{C}} \kappa(v) = n - 1$, this implies $\mathrm{trdeg}_{\mathbf{C}} \kappa(w) \geq m - 1$. The Abhyankar inequality gives the reverse inequality. Thus w is divisorial. Finitely many such restrictions can be extracted on one smooth projective birational model of I .

Only finitely many prime divisors of the fixed H have image of codimension at least two in I : they lie in the complement of a locus where p is smooth and are components of its divisorial part. Extract their restrictions as above. Original horizontal primes remain horizontal, while original vertical primes then have divisorial centers on the prepared base. These centers remain divisorial on higher base models. Hence on any subsequent commuting birational diagram, a source prime whose base image still has codimension at least two is exceptional over the fixed H .

For the rank-one section comparison, fix the original smooth pair (X, D_X) , and let $X' \rightarrow X$ carry strict plus reduced exceptional boundary D' . Let $x : X' \rightarrow W$ be the relative Iitaka fibration on smooth projective models; its geometric generic fiber (J, D_J) has logarithmic Kodaira dimension zero. Let s be a nonzero relative rational p -canonical form generating the degree- p generic section space. For a source prime E , put $d_E = \mathrm{coeff}_E D'$, and for a base prime P define

$$t_P = \min_{E \mapsto P} \frac{p^{-1} \mathrm{ord}_E^{pK_{X'/W}}(s) + d_E}{\mathrm{ord}_E(x^*P)}, \quad T_W = \sum_P t_P P.$$

Here T_W is the entire minimum-rule base term, before its division into discriminant and moduli parts. In divisible degrees, division by $s^{m/p}$ gives

$$(A.1) \quad H^0(X', m(K_{X'} + D')) \hookrightarrow H^0(W, m(K_W + T_W)).$$

Indeed, on the geometric generic fiber the section is a constant multiple of $s^{m/p}$; connectedness identifies that coefficient with a rational m -canonical form σ on W . At a component E dominating P , write σ in a nonvanishing local frame of mK_W , and put $a_E = \mathrm{ord}_E(x^*P)$. Its logarithmic regularity upstairs is equivalent to

$$a_E \mathrm{ord}_P(\sigma) + (m/p) \mathrm{ord}_E^{pK_{X'/W}}(s) + md_E \geq 0.$$

The inequalities for all components above P are exactly $\mathrm{ord}_P(\sigma) + mt_P \geq 0$. Horizontal primes impose no further condition because s is an allowed generic logarithmic pluriform. The same argument works over the generic point of the original base Y . If the relative Iitaka dimension is k , then $\dim(W/Y) = k$, so preservation of section ratios makes $K_W + T_W$ big over Y .

For the converse, extract on W the restrictions of all source divisors with exceptional image that survive as divisors on the original X . Resolve the induced diagram and recompute the minimum rule. Every divisor of the original X is now tested either by the generic horizontal condition or by a base prime. A section on the right of (A.1), multiplied by $s^{m/p}$, therefore satisfies the logarithmic order inequality at every original source prime. Any remaining untested

pole lies on a divisor exceptional over X . The codimension-one criterion on the normal original source gives a section of $m(K_X + D_X)$; later models preserve this criterion. Conversely the logarithmic modification identity recovers its section on X' . The identifications respect powers and multiplication. Under these identifications, corresponding section ratios agree after pullback by $x : X' \rightarrow W$. Their section fields and rational-map image dimensions therefore agree.

A.3. The fixed zero divisor and the discrepancy-sheaf test. In the notation of Lemma 7.1, the rational divisor $Z = E/p$ is independent of the chosen nonzero logarithmic pluriform and its degree. Indeed, if ω_m is a nonzero section in another degree m , the degree- mp sections ω_m^p and ω^m are proportional, since that section space is one-dimensional. Consequently

$$p(\operatorname{div}(\omega_m) + mD) = mE.$$

The least positive integer trivializing $K_J + D - Z$ is also p . Writing $\omega = b\eta^{\otimes p}$ in a rational canonical frame gives $p(K_J + D - Z) = -\operatorname{div}(b)$. Conversely, a trivialization of $r(K_J + D - Z)$ gives a rational r -canonical form whose logarithmic zero divisor is $rZ \geq 0$, hence a nonzero logarithmic pluriform of degree r . Minimality gives $r \geq p$.

Here is a second way to check the generic discrepancy-sheaf rank condition in Corollary 7.2. Take a log resolution $J' \rightarrow J$, let $D_{J'}$ be the strict boundary plus the reduced exceptional divisor, and put

$$E_{J'} = \operatorname{div}(\omega|_{J'}) + pD_{J'} \geq 0.$$

The crepant transform of the auxiliary generic subboundary is $D_{J'} - E_{J'}/p$. Its coefficients are at most one. If A^* is the discrepancy b-divisor with coefficient -1 replaced by zero, the rounded trace is effective, and a positive coefficient can occur only on $\operatorname{Supp} E_{J'}$. Coefficientwise,

$$0 \leq \lceil A^* \rceil_{J'} \leq \lceil E_{J'}/p \rceil \leq E_{J'}.$$

The first inequality uses the omission of discrepancy -1 . Logarithmic birational invariance gives $\kappa(E_{J'}) = 0$. The associated space of rational functions therefore has dimension at most one and contains the constants. This proves rank one on every sufficiently high resolution, including when the auxiliary subboundary has negative coefficients.

A.4. A toroidal proof of logarithmic extension. The extension part of Lemma 7.1 also has a direct coordinate proof on a toroidal resolution of the SNC cyclic-cover construction. First resolve the supports on J , retaining strict plus reduced exceptional logarithmic boundary. The minimal index and the fixed-zero assertions are unchanged. In a local SNC chart write

$$\omega = \varepsilon \prod_i z_i^{a_i - pd_i} (dz_1 \wedge \cdots \wedge dz_r)^{\otimes p}, \quad a_i \geq 0, \quad d_i \in \{0, 1\},$$

where ε is a unit. Include all coordinates; outside $D + \operatorname{Supp} E$ put $d_i = a_i = 0$. After a local unramified change the unit has a root and contributes no divisorial order. Let v be a toroidal prime on the resolved root cover, and put $n_i = v(z_i) \geq 0$. The nonzero ray defining this prime has at least one positive n_i . Rewriting with logarithmic differentials gives

$$(A.2) \quad \operatorname{ord}_v(\Omega) = \sum_i n_i \left(1 - d_i + \frac{a_i}{p} \right) - 1.$$

The logarithmic volume form has order -1 along a toroidal prime, and the change of torus lattice multiplies it by a nonzero constant. Every coefficient in the sum is nonnegative. It vanishes exactly when $d_i = 1$ and $a_i = 0$, namely along the coefficient-one pole boundary of the auxiliary subpair.

If v is outside the inverse image of this pole boundary, then $n_i = 0$ for each such zero coefficient. Some remaining n_i is positive, so the sum in (A.2) is strictly positive. The order of the rational top form Ω is integral, so that positive sum is at least one, giving regularity. On the inverse

pole boundary the nonnegative sum gives order at least -1 . Primes in the smooth unramified locus have the regularity already checked on the finite normalization. Thus Ω extends as a top form with logarithmic poles allowed only along the inverse pole boundary. This provides the coordinate alternative to the discrepancy-scaling proof.

A.5. The normalized discriminant on higher models. Let $x : X' \rightarrow W$ and D' be the projective reduced-boundary data of Lemma 7.4. Write s for its relative rational p -canonical generator, and set

$$t_P = \min_{E \mapsto P} \frac{p^{-1} \operatorname{ord}_E^{pK_{X'/W}}(s) + d_E}{\operatorname{ord}_E(x^*P)}, \quad T_W = \sum_P t_P P,$$

$$\Delta_0 = -p^{-1} \operatorname{div}_{K_{X'/W}}(s), \quad \Delta = \Delta_0 + x^*T_W.$$

In this subsection T_W denotes the entire minimum-rule base term, before its decomposition into discriminant and moduli parts; the boundary denoted T in Section 2.6 is the discriminant part. We have $K_{X'} + \Delta \sim_{\mathbf{Q}} x^*(K_W + T_W)$. The auxiliary divisor is required to be sub-lc over the generic point of W . Its coefficients at source divisors with image of codimension at least two are not asserted to be at most one.

For a prime $P \subset W$, let

$$c_P = \operatorname{lc}_{\eta_P}(X', \Delta; x^*P), \quad b_P = 1 - c_P, \quad B_W = \sum_P b_P P.$$

Over the generic point of P the minimum rule gives $\Delta \leq D'$, including the horizontal divisors. Log canonicity therefore gives $c_P \geq 0$. A component E attaining the minimum has coefficient exactly $d_E \in \{0, 1\}$ in Δ , and hence

$$c_P \leq \frac{1 - d_E}{\operatorname{ord}_E(x^*P)} \leq 1.$$

It follows that $0 \leq B_W \leq 1$. If P lies in a required inverse boundary from a base Y , all components over P have $d_E = 1$, so $c_P = 0$ and $b_P = 1$.

For the exact model comparison, let $\beta : W_1 \rightarrow W_0$ be a higher smooth base model and let $\pi : X_1 \rightarrow X_0$ resolve the induced diagram, with $x_i : X_i \rightarrow W_i$. Choose compatible rational top forms for the canonical divisors. First form the crepant transform

$$K_{X_1} + \Delta_{\text{crepant}} = \pi^*(K_{X_0} + \Delta_{\text{old}}).$$

Using the new logarithmic boundary and the same relative trivialization s , recompute the minimum-rule divisor T_{W_1} and the normalized auxiliary divisor Δ_{new} . Comparison of their definitions gives

$$(A.3) \quad \begin{aligned} Q &= K_{W_1} + T_{W_1} - \beta^*(K_{W_0} + T_{W_0}), \\ \Delta_{\text{new}} &= \Delta_{\text{crepant}} + x_1^*Q. \end{aligned}$$

Thus the twist includes the canonical discrepancy of the base. Adding x^*Q lowers the threshold at P by $\operatorname{coeff}_P Q$, adding Q both to the discriminant and to the total base term. It leaves the moduli part unchanged. Birational pullback of the lc-trivial fibration also preserves its moduli b-divisor; these compatibilities agree with the Hodge extension description in [7, Remarks 6.22 and 6.25].

Resolve the finite supports on W and resolve the induced source diagram. At an unchanged codimension-one point, logarithmic forms satisfying the old tests remain logarithmic on the source resolution, while the strict old minimizing components persist. Thus the minimum and discriminant trace there are unchanged. New support is contained in the base-exceptional divisor. Its union with the old strict support is SNC on the chosen resolution; recomputing and applying

the preceding coefficient argument yields an effective SNC discriminant with the same coefficient-one inverse boundary and the same moduli b -divisor.

Finally, if $m = ap$ and $s_m = cs^a$ on the generic fiber, with $c \in \mathbf{C}(W)^*$, the minimum construction gives

$$T_W(s_m) = T_W(s) + m^{-1} \operatorname{div}(c).$$

The two occurrences of $\operatorname{div}(c)$ cancel in the definition of the normalized auxiliary subpair. It is unchanged. The minimal-index cover is therefore compatible with all common divisible degrees used for section descent.

REFERENCES

- [1] Karim Adiprasito, Gaku Liu and Michael Temkin. *Semistable reduction in characteristic 0*. Preprint, 2019. arXiv:1810.03131v2, 17 June 2019; 22-page full preprint.
- [2] Florin Ambro. *Shokurov’s boundary property*. Journal of Differential Geometry **67** (2004), no. 2, 229–255. Theorem numbering is cited from arXiv:math/0210271v1.
- [3] Florin Ambro. *The moduli b -divisor of an lc-trivial fibration*. Compositio Mathematica **141** (2005), no. 2, 385–403. doi:10.1112/S0010437X04001071.
- [4] Yves André. *Mumford–Tate groups of mixed Hodge structures and the theorem of the fixed part*. Compositio Mathematica **82** (1992), no. 1, 1–24.
- [5] Benjamin Bakker. *A short proof of a conjecture of Matsushita*. Advances in Mathematics **482** (2025), 110554. doi:10.1016/j.aim.2025.110554. Part A.
- [6] Benjamin Bakker, Yohan Brunebarbe and Jacob Tsimerman. *o -minimal GAGA and a conjecture of Griffiths*. Inventiones Mathematicae **232** (2023), no. 1, 163–228. doi:10.1007/s00222-022-01166-1. Theorem locators refer to arXiv:1811.12230v3, 23 October 2022.
- [7] Benjamin Bakker, Stefano Filipazzi, Mirko Mauri and Jacob Tsimerman. *Baily–Borel compactifications of period images and the b -semiample conjecture*. Preprint, 2025. arXiv:2508.19215v2, 18 December 2025.
- [8] Benjamin Bakker and Jacob Tsimerman. *The Ax–Schanuel conjecture for variations of Hodge structures*. Inventiones Mathematicae **217** (2019), no. 1, 77–94. doi:10.1007/s00222-019-00863-8. The cited locators refer to arXiv:1712.05088v1.
- [9] Edward Bierstone and Pierre D. Milman. *Functoriality in resolution of singularities*. Publications of the Research Institute for Mathematical Sciences **44** (2008), no. 2, 609–639. doi:10.2977/PRIMS/1210167338.
- [10] Caucher Birkar. *The Iitaka conjecture $C_{n,m}$ in dimension six*. Compositio Mathematica **145** (2009), no. 6, 1442–1446. doi:10.1112/S0010437X09004187. arXiv:0806.4413, first posted 27 June 2008.
- [11] Caucher Birkar, Paolo Cascini, Christopher D. Hacon and James McKernan. *Existence of minimal models for varieties of log general type*. Journal of the American Mathematical Society **23** (2010), no. 2, 405–468. doi:10.1090/S0894-0347-09-00649-3. The cited locators refer to the 81-page arXiv:math/0610203v2.
- [12] Sébastien Boucksom. *On the volume of a line bundle*. International Journal of Mathematics **13** (2002), no. 10, 1043–1063. doi:10.1142/S0129167X02001575. Theorem locators refer to arXiv:math/0201031v1, 5 January 2002.
- [13] Sébastien Boucksom, Philippe Eyssidieux, Vincent Guedj and Ahmed Zeriahi. *Monge–Ampère equations in big cohomology classes*. Acta Mathematica **205** (2010), 199–262. doi:10.1007/s11511-010-0054-7.
- [14] Yohan Brunebarbe and Benoît Cadorel. *Hyperbolicity of varieties supporting a variation of Hodge structure*. International Mathematics Research Notices **2020** (2020), no. 6, 1601–1609. doi:10.1093/imrn/rny054. The cited locators refer to arXiv:1707.01327v1.
- [15] Frédéric Campana. *Orbifolds, special varieties and classification theory*. Annales de l’Institut Fourier **54** (2004), no. 3, 499–630. doi:10.5802/aif.2027.
- [16] Frédéric Campana. *Orbifolds géométriques spéciales et classification biméromorphe des variétés kählériennes compactes*. Journal of the Institute of Mathematics of Jussieu **10** (2011), no. 4, 809–934. doi:10.1017/S1474748010000101. The cited locators refer to arXiv:0705.0737v8.
- [17] Junyan Cao. *Kodaira dimension of algebraic fiber spaces over surfaces*. Algebraic Geometry **5** (2018), no. 6, 728–741. doi:10.14231/AG-2018-020.
- [18] Junyan Cao and Mihai Păun. *Kodaira dimension of algebraic fiber spaces over abelian varieties*. Inventiones Mathematicae **207** (2017), no. 1, 345–387. doi:10.1007/s00222-016-0672-6. Theorem locators refer to arXiv:1504.01095v3, submitted 3 August 2016.
- [19] Eduardo Cattani, Aroldo Kaplan and Wilfried Schmid. *Degeneration of Hodge structures*. Annals of Mathematics (2) **123** (1986), no. 3, 457–535. doi:10.2307/1971333.

- [20] Qianyu Chen. *Limits of Hodge structures via holonomic D -modules*. Preprint, 2021. arXiv:2103.03983; the cited Lemma 8.1 is in the 61-page author manuscript.
- [21] Pierre Deligne. *Théorie de Hodge. II*. Publications Mathématiques de l’IHÉS **40** (1971), 5–57. doi:10.1007/BF02684692.
- [22] Jean-Pierre Demailly. *Regularization of closed positive currents and intersection theory*. Journal of Algebraic Geometry **1** (1992), no. 3, 361–409. The cited locator refers to the author manuscript.
- [23] Jean-Pierre Demailly and Mihai Păun. *Numerical characterization of the Kähler cone of a compact Kähler manifold*. Annals of Mathematics (2) **159** (2004), no. 3, 1247–1274. doi:10.4007/annals.2004.159.1247.
- [24] Adrien Douady. *Le problème des modules pour les sous-espaces analytiques compacts d’un espace analytique donné*. Annales de l’Institut Fourier **16** (1966), no. 1, 1–95. doi:10.5802/aif.226.
- [25] Makoto Enokizono and Kenta Hashizume. *Semistable reduction for complex analytic spaces*. Transactions of the American Mathematical Society **378** (2025), 7667–7688. doi:10.1090/tran/9365. The cited locators refer to arXiv:2404.05124v3, 12 October 2024.
- [26] Akira Fujiki. *Closedness of the Douady spaces of compact Kähler spaces*. Publications of the Research Institute for Mathematical Sciences **14** (1978), no. 1, 1–52. doi:10.2977/PRIMS/1195189279.
- [27] Osamu Fujino. *Notes on the weak positivity theorems*. In *Algebraic Varieties and Automorphism Groups*. Advanced Studies in Pure Mathematics 75. Mathematical Society of Japan, 2017, pp. 73–118. doi:10.2969/aspm/07510073. The cited locators refer to author version 0.54, 30 June 2015.
- [28] Osamu Fujino. *Log canonical inversion of adjunction*. Proceedings of the Japan Academy, Series A, Mathematical Sciences **100** (2024), no. 2, 7–11. doi:10.3792/PJAA.100.002.
- [29] Osamu Fujino and Taro Fujisawa. *Variations of mixed Hodge structure and semipositivity theorems*. Publications of the Research Institute for Mathematical Sciences **50** (2014), no. 4, 589–661. doi:10.4171/PRIMS/145.
- [30] Osamu Fujino and Taro Fujisawa. *Variation of mixed Hodge structure and its applications*. International Journal of Mathematics **36** (2025), no. 9, 2550020. Published article: 43 pp.; theorem locators refer to the 36-page author version 0.55, 11 March 2025.
- [31] Osamu Fujino and Yoshinori Gongyo. *On the moduli b -divisors of lc-trivial fibrations*. Annales de l’Institut Fourier **64** (2014), no. 4, 1721–1735. doi:10.5802/aif.2894.
- [32] Osamu Fujino and Shigefumi Mori. *A canonical bundle formula*. Journal of Differential Geometry **56** (2000), no. 1, 167–188.
- [33] Hans Grauert. *Ein Theorem der analytischen Garbentheorie und die Modulräume komplexer Strukturen*. Publications Mathématiques de l’IHÉS **5** (1960), 5–64. doi:10.1007/BF02684746. With corrigendum, Publ. Math. IHÉS **16** (1963), 35–36.
- [34] Hans Grauert. *Berichtigung: Ein Theorem der analytischen Garbentheorie und die Modulräume komplexer Strukturen*. Publications Mathématiques de l’IHÉS **16** (1963), 35–36. doi:10.1007/BF02684291.
- [35] Phillip A. Griffiths. *Periods of integrals on algebraic manifolds, III: Some global differential-geometric properties of the period mapping*. Publications Mathématiques de l’IHÉS **38** (1970), 125–180. doi:10.1007/BF02684654.
- [36] Vincent Guedj and Ahmed Zeriahi. *The weighted Monge–Ampère energy of quasisubharmonic functions*. Journal of Functional Analysis **250** (2007), no. 2, 442–482. doi:10.1016/j.jfa.2007.04.018. The cited locators refer to arXiv:math/0612630v2.
- [37] Christopher D. Hacon, Mihnea Popa and Christian Schnell. *Algebraic fiber spaces over abelian varieties: around a recent theorem by Cao and Păun*. In *Local and Global Methods in Algebraic Geometry*. Contemporary Mathematics 712. American Mathematical Society, Providence, RI, 2018, pp. 143–195. doi:10.1090/conm/712/14346. Theorem locators refer to the author manuscript dated 9 May 2017.
- [38] Kenta Hashizume. *Log Iitaka conjecture for abundant log canonical fibrations*. Proceedings of the Japan Academy, Series A, Mathematical Sciences **96** (2020), no. 10, 87–92. doi:10.3792/pjaa.96.017. Theorem numbering is cited from arXiv:1902.10923v2.
- [39] Heisuke Hironaka. *Flattening theorem in complex-analytic geometry*. American Journal of Mathematics **97** (1975), no. 2, 503–547. doi:10.2307/2373721.
- [40] Shigeru Iitaka. *On D -dimensions of algebraic varieties*. Journal of the Mathematical Society of Japan **23** (1971), no. 2, 356–373. doi:10.2969/jmsj/02320356.
- [41] Shigeru Iitaka. *On logarithmic Kodaira dimension of algebraic varieties*. In *Complex Analysis and Algebraic Geometry: A Collection of Papers Dedicated to K. Kodaira*. Walter L. Baily, Jr. and Tetsuji Shioda (eds.). Iwanami Shoten and Cambridge University Press, 1977. doi:10.1017/CB09780511569197.014.
- [42] Shigeru Iitaka. *Classification of Algebraic Varieties*. Proceedings of the Japan Academy **53** (1977), no. 3, 103–105. doi:10.3792/pja/1195517881.

- [43] Yujiro Kawamata. *Kodaira dimension of algebraic fiber spaces over curves*. *Inventiones Mathematicae* **66** (1982), no. 1, 57–71. doi:10.1007/BF01404756.
- [44] Yujiro Kawamata. *Minimal models and the Kodaira dimension of algebraic fiber spaces*. *Journal für die reine und angewandte Mathematik* **363** (1985), 1–46. doi:10.1515/crll.1985.363.1.
- [45] János Kollár. *Subadditivity of the Kodaira dimension: fibers of general type*. In *Algebraic Geometry, Sendai, 1985*. Advanced Studies in Pure Mathematics 10. Kinokuniya and North-Holland, Tokyo and Amsterdam, 1987, pp. 361–398. doi:10.2969/aspm/01010361.
- [46] Sándor J. Kovács and Zolt Patakfalvi. *Projectivity of the moduli space of stable log-varieties and subadditivity of log-Kodaira dimension*. *Journal of the American Mathematical Society* **30** (2017), no. 4, 959–1021. doi:10.1090/jams/871. The cited locators refer to the 62-page author manuscript.
- [47] Kazuhisa Maehara. *Iitaka–Viehweg conjectures C and C^{++}* . arXiv preprint, 2010. Version 3, 16 May 2010; first posted 4 October 2009.
- [48] James S. Milne. *Algebraic Groups: The Theory of Group Schemes of Finite Type over a Field*. Cambridge Studies in Advanced Mathematics 170. Cambridge University Press, 2017. doi:10.1017/9781316711736. Cited from the corrected 2022 reprint.
- [49] Yoshinori Namikawa. *Projectivity criterion of Moishezon spaces and density of projective symplectic varieties*. *International Journal of Mathematics* **13** (2002), no. 2, 125–135. Theorem locators refer to arXiv:math/0101019v6, 12 November 2001.
- [50] OpenAI. *Logarithmic Kodaira dimension and whole-fiber variation*. OpenAI Math Release preprint OAI:Logarithmic-Kodaira-dimension-and-whole-fiber-variation-September-26-2026, 2026.
- [51] OpenAI. *Projective Hodge lines and ordinary Iitaka subadditivity*. OpenAI Math Release preprint OAI:Projective-Hodge-lines-and-ordinary-Iitaka-subadditivity-September-27-2026, 2026.
- [52] OpenAI. *The reverse logarithmic Kodaira inequality and additivity*. OpenAI Math Release preprint OAI:The-reverse-logarithmic-Kodaira-inequality-and-additivity-September-26-2026, 2026.
- [53] OpenAI. *Log abundance in characteristic zero*. OpenAI Math Release preprint OAI:Log-abundance-in-characteristic-zero-September-24-2026, 2026.
- [54] Giuseppe Pareschi, Mihnea Popa and Christian Schnell. *Hodge modules on complex tori and generic vanishing for compact Kähler manifolds*. *Geometry & Topology* **21** (2017), 2419–2460. doi:10.2140/gt.2017.21.2419.
- [55] Zolt Patakfalvi. *Fibered stable varieties*. *Transactions of the American Mathematical Society* **368** (2016), no. 3, 1837–1869. doi:10.1090/tran/6386.
- [56] Wilfried Schmid. *Variation of Hodge structure: The singularities of the period mapping*. *Inventiones Mathematicae* **22** (1973), no. 3–4, 211–319. doi:10.1007/BF01389674.
- [57] Andrew John Sommese. *On the rationality of the period mapping*. *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze* (4) **5** (1978), no. 4, 683–717.
- [58] Hajime Tsuji. *Global generation of the direct images of relative pluricanonical systems*. arXiv preprint, 2010. Version 1, 4 December 2010.
- [59] Kenji Ueno. *Classification Theory of Algebraic Varieties and Compact Complex Spaces*. Lecture Notes in Mathematics 439. Springer, Berlin, 1975. doi:10.1007/BFb0070570.
- [60] Eckart Viehweg. *Weak positivity and the additivity of the Kodaira dimension for certain fibre spaces*. In *Algebraic Varieties and Analytic Varieties (Tokyo, 1981)*. Advanced Studies in Pure Mathematics 1. Kinokuniya and North-Holland, Tokyo and Amsterdam, 1983, pp. 329–353. doi:10.2969/aspm/00110329.
- [61] Mads Bach Villadsen. *Hodge modules and Kähler morphisms*. Preprint, 2024. arXiv:2401.09544v1, 17 January 2024.
- [62] Juanyong Wang. *On the Iitaka Conjecture $C_{n,m}$ for Kähler Fibre Spaces*. *Annales de la Faculté des Sciences de Toulouse. Mathématiques* (6) **30** (2021), no. 4, 813–897. doi:10.5802/afst.1690.